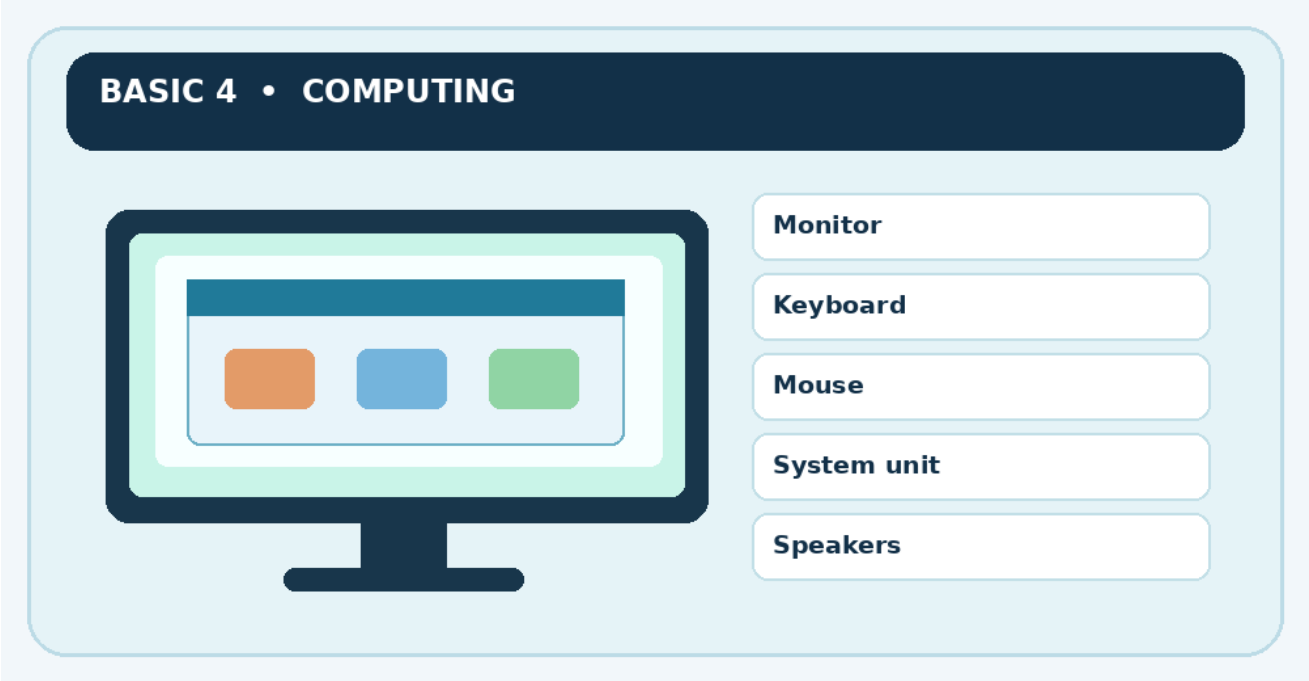
COMPUTING

Basic 4 — Illustrated Learning Workbook

Learn • Explore • Practise • Play • Challenge



27 lessons | 135 Play questions | 270 Challenge questions

Basic 4–6: independent adaptation of selected indicators in the 2019 NaCCA Computing curriculum; not an exhaustive mapping.

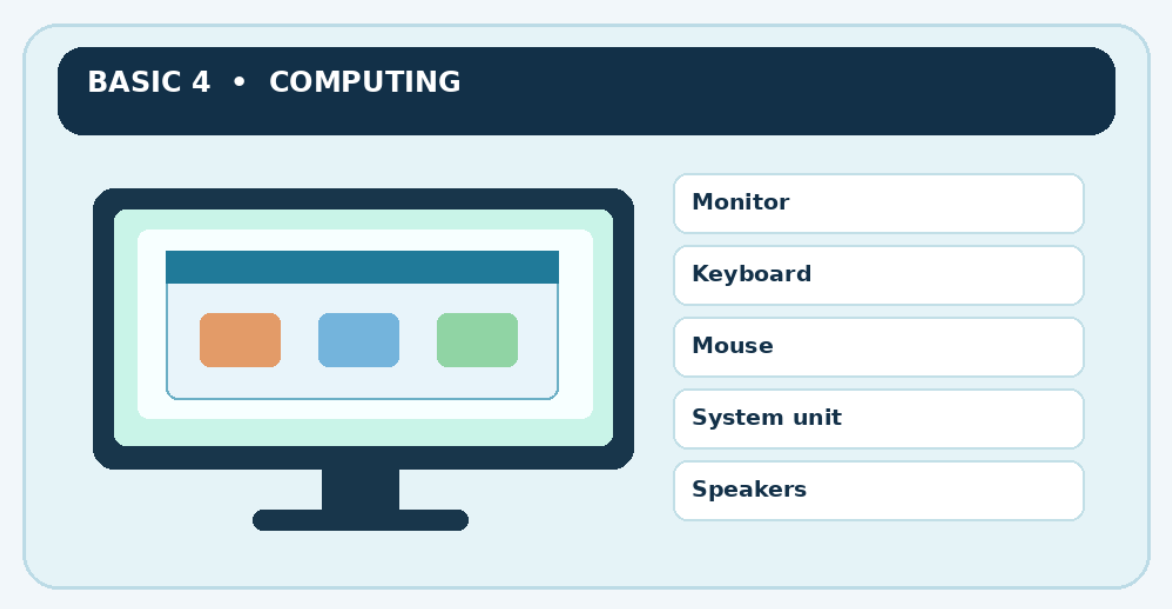
Course guide and lesson list

Start with the picture and five important ideas. Read the Ghanaian example, complete the practice activity and answer the quizzes. An adult should supervise device use and online activities.

01. Computer Components — Computing Systems
02. Different Input Devices — Computing Systems
03. RAM and ROM — Computing Systems
04. Typing Short Phrases — Computing Systems
05. First Generation Computers — Computing Systems
06. Desktop Features — Computer Applications
07. Personalise the Desktop — Computer Applications
08. Taskbar and Start Menu — Computer Applications
09. Add and Remove Desktop Icons — Computer Applications
10. Moving and Selecting Icons — Computing Systems
11. Creating and Renaming Folders — Computer Applications
12. Identify Data in Everyday Reports — Data and Information
13. Classify Data into Information — Data and Information
14. Recognising Information Sources — Data and Information
15. Sharing Information With Devices — Internet and Communication
16. Calculating With Simple Data — Data and Information
17. Steps in Data Collection — Data and Information
18. Data Collection Tools — Data and Information
19. Sketch a Simple Community Map — Data and Information
20. Digital Tools in the Community — Technology in Society
21. Energy-Saving Technology — Technology in Society
22. Digital Devices and Peripherals — Computing Systems
23. Changing Technology Around Us — Technology in Society
24. PowerPoint Home Tools — Presentation
25. Make a Three-Slide Presentation — Presentation
26. Word Processor Home Tools — Word Processing
27. Format a Simple Document — Word Processing

Lesson 1: Computer Components

We will learn to: Identify, explain and apply the important ideas behind computer components.



Five important ideas

Monitor: shows words and pictures.

Keyboard: enters letters and numbers.

Mouse: moves and selects a pointer.

System unit: contains important computing components.

Speakers: play sound from a computer.

Our Ghanaian story

At the school ICT laboratory, Kojo had a question about computer components. The group discovered that monitor shows words and pictures, while keyboard enters letters and numbers. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of monitor, keyboard and mouse. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches computer components. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.1.1.1

Lesson 1: Play & Challenge

PLAY — Choose one best answer for each question.

1. Computer Components — Choose the correct name for something that shows words and pictures.

A. Mouse B. Keyboard C. Speakers D. Monitor

2. Computer Components — Your friend says, "This enters letters and numbers." What is the friend describing?

A. Mouse B. System unit C. Speakers D. Keyboard

3. Computer Components — Which idea from computer components matches the clue "moves and selects a pointer"?

A. Speakers B. Mouse C. System unit D. Keyboard

4. Computer Components — The teacher asks for an example that contains important computing components. Which answer fits?

A. Keyboard B. Monitor C. System unit D. Speakers

5. Computer Components — Can you identify the term described here: play sound from a computer?

A. Keyboard B. System unit C. Speakers D. Monitor

CHALLENGE — Think and apply.

1. Computer Components — At home, Yaw remembers a class rule or concept that contains important computing components. What was it called?

A. Keyboard B. Monitor C. System unit D. Speakers

2. Computer Components — A group makes a display on computer components. Their first clue is "enters letters and numbers". Which label is correct?

A. Speakers B. Keyboard C. System unit D. Mouse

3. Computer Components — For a short demonstration, the teacher needs an example that moves and selects a pointer. Which choice would you select?

A. Speakers B. Mouse C. System unit D. Keyboard

4. Computer Components — Which concept matches the explanation "play sound from a computer" when you review computer components?

A. Speakers B. Monitor C. Keyboard D. System unit

5. Computer Components — At the community learning centre, someone describes a tool or idea that play sound from a computer. Which one is it?

A. Monitor B. System unit C. Speakers D. Keyboard

6. Computer Components — Your class is solving a practical problem. One useful thing contains important computing components. What is its name?

A. Keyboard B. Speakers C. Monitor D. System unit

7. Computer Components — Which answer would fit on a poster beside the words "shows words and pictures"?

A. Mouse B. Keyboard C. Speakers D. Monitor

8. Computer Components — During a school ICT lesson, Yaw wants to show something that shows words and pictures. Which idea should be labelled?

A. Mouse B. Speakers C. Keyboard D. Monitor

9. Computer Components — Imagine helping a younger pupil understand something that moves and selects a pointer. Which term would you teach?

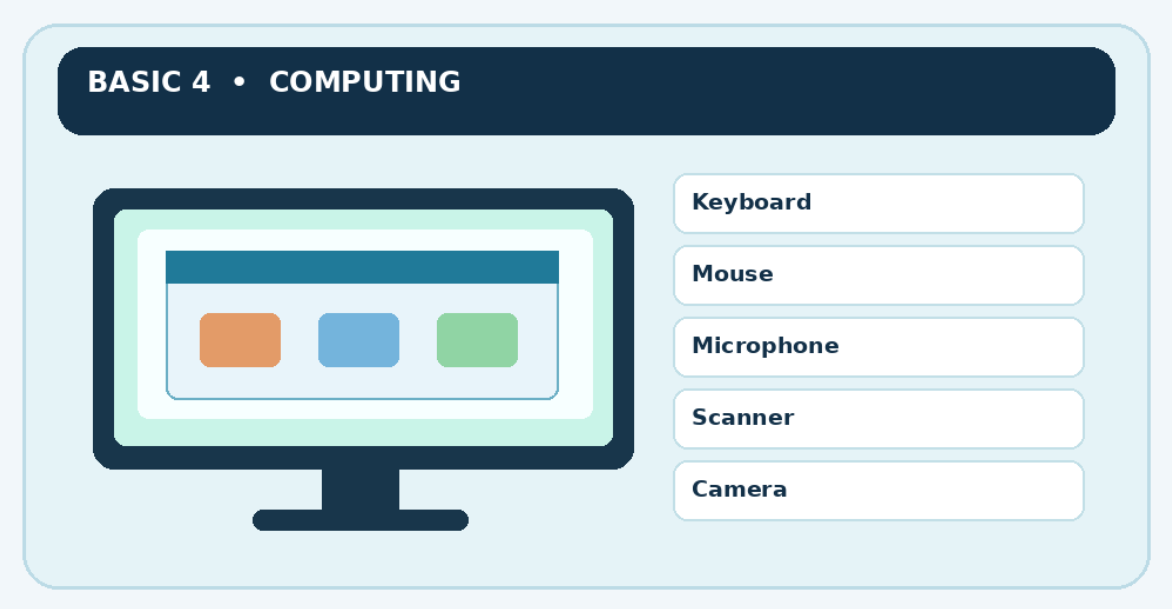
A. Mouse B. System unit C. Keyboard D. Speakers

10. Computer Components — A classmate confuses several ideas about computer components. Which one actually enters letters and numbers?

A. Speakers B. System unit C. Keyboard D. Mouse

Lesson 2: Different Input Devices

We will learn to: Identify, explain and apply the important ideas behind different input devices.



Five important ideas

Keyboard: inputs typed letters and digits.

Mouse: inputs clicks and movements.

Microphone: inputs recorded sound.

Scanner: inputs pictures from paper.

Camera: inputs digital photographs.

Our Ghanaian story

When the class records a school event, Esi had a question about different input devices. The group discovered that keyboard inputs typed letters and digits, while mouse inputs clicks and movements. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of keyboard, mouse and microphone. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches different input devices. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.1.1.2

Lesson 2: Play & Challenge

PLAY — Choose one best answer for each question.

1. Different Input Devices — Choose the correct name for something that inputs typed letters and digits.

A. Camera B. Scanner C. Keyboard D. Microphone

2. Different Input Devices — Your friend says, "This inputs clicks and movements." What is the friend describing?

A. Mouse B. Microphone C. Camera D. Scanner

3. Different Input Devices — Which idea from different input devices matches the clue "inputs recorded sound"?

A. Mouse B. Scanner C. Microphone D. Keyboard

4. Different Input Devices — The teacher asks for an example that inputs pictures from paper. Which answer fits?

A. Mouse B. Camera C. Microphone D. Scanner

5. Different Input Devices — Can you identify the term described here: inputs digital photographs?

A. Camera B. Scanner C. Microphone D. Keyboard

CHALLENGE — Think and apply.

1. Different Input Devices — At the community learning centre, someone describes a tool or idea that inputs digital photographs. Which one is it?

A. Scanner B. Camera C. Microphone D. Keyboard

2. Different Input Devices — A group makes a display on different input devices. Their first clue is "inputs clicks and movements". Which label is correct?

A. Microphone B. Mouse C. Camera D. Scanner

3. Different Input Devices — During a school ICT lesson, Kofi wants to show something that inputs typed letters and digits. Which idea should be labelled?

A. Keyboard B. Scanner C. Camera D. Microphone

4. Different Input Devices — Which concept matches the explanation "inputs digital photographs" when you review different input devices?

A. Microphone B. Scanner C. Camera D. Keyboard

5. Different Input Devices — For a short demonstration, the teacher needs an example that inputs recorded sound. Which choice would you select?

A. Microphone B. Mouse C. Scanner D. Keyboard

6. Different Input Devices — A classmate confuses several ideas about different input devices. Which one actually inputs clicks and movements?

A. Microphone B. Mouse C. Camera D. Scanner

7. Different Input Devices — Imagine helping a younger pupil understand something that inputs recorded sound. Which term would you teach?

A. Keyboard B. Microphone C. Scanner D. Mouse

8. Different Input Devices — Which answer would fit on a poster beside the words "inputs typed letters and digits"?

A. Camera B. Scanner C. Keyboard D. Microphone

9. Different Input Devices — At home, Kofi remembers a class rule or concept that inputs pictures from paper. What was it called?

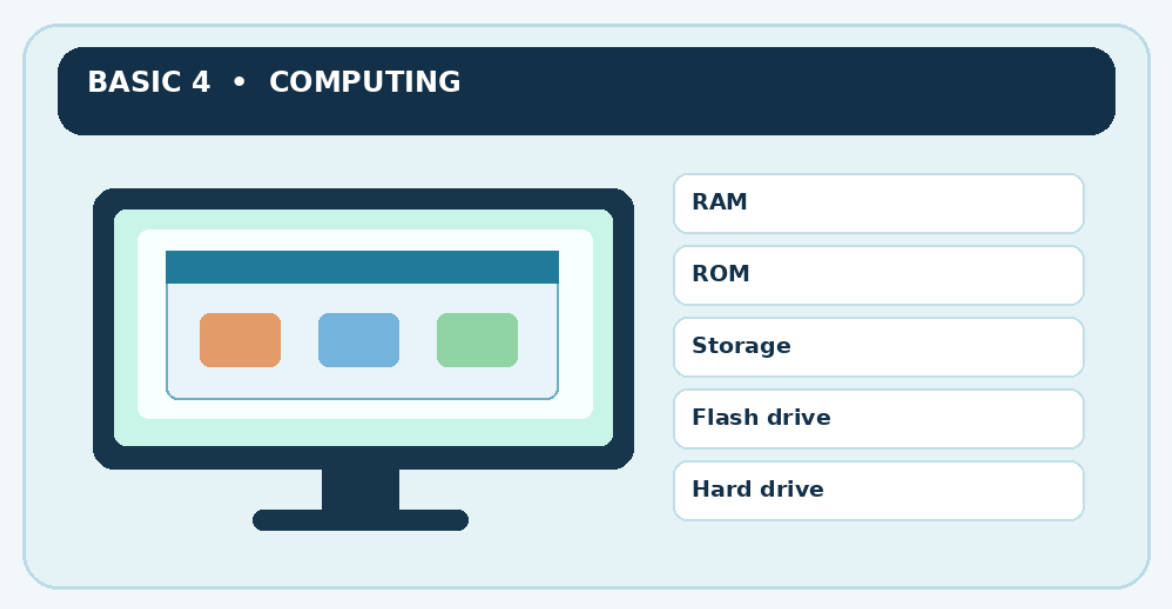
A. Scanner B. Camera C. Mouse D. Microphone

10. Different Input Devices — Your class is solving a practical problem. One useful thing inputs pictures from paper. What is its name?

A. Microphone B. Scanner C. Camera D. Mouse

Lesson 3: RAM and ROM

We will learn to: Identify, explain and apply the important ideas behind ram and rom.



Five important ideas

RAM: is temporary working memory used while programs run.

ROM: stores instructions that normally remain when power is off.

Storage: keeps files for later use.

Flash drive: is portable file storage.

Hard drive: is a storage device used in many computers.

Our Ghanaian story

While comparing a computer with a notebook, Yaw had a question about ram and rom. The group discovered that ram is temporary working memory used while programs run, while rom stores instructions that normally remain when power is off. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of ram, rom and storage. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches ram and rom. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.1.1.3

Lesson 3: Play & Challenge

PLAY — Choose one best answer for each question.

1. RAM and ROM — Choose the correct name for something that is temporary working memory used while programs run.

A. RAM B. ROM C. Storage D. Hard drive

2. RAM and ROM — Your friend says, "This stores instructions that normally remain when power is off." What is the friend describing?

A. Hard drive B. Storage C. Flash drive D. ROM

3. RAM and ROM — Which idea from ram and rom matches the clue "keeps files for later use"?

A. RAM B. ROM C. Hard drive D. Storage

4. RAM and ROM — The teacher asks for an example that is portable file storage. Which answer fits?

A. ROM B. RAM C. Hard drive D. Flash drive

5. RAM and ROM — Can you identify the term described here: is a storage device used in many computers?

A. RAM B. Hard drive C. ROM D. Storage

CHALLENGE — Think and apply.

1. RAM and ROM — A classmate confuses several ideas about ram and rom. Which one actually stores instructions that normally remain when power is off?

A. Storage B. Flash drive C. Hard drive D. ROM

2. RAM and ROM — Your class is solving a practical problem. One useful thing is portable file storage. What is its name?

A. Flash drive B. Hard drive C. RAM D. ROM

3. RAM and ROM — A group makes a display on ram and rom. Their first clue is "stores instructions that normally remain when power is off". Which label is correct?

A. ROM B. Hard drive C. Flash drive D. Storage

4. RAM and ROM — During a school ICT lesson, Akosua wants to show something that is temporary working memory used while programs run. Which idea should be labelled?

A. Hard drive B. RAM C. ROM D. Storage

5. RAM and ROM — At the community learning centre, someone describes a tool or idea that is a storage device used in many computers. Which one is it?

A. Storage B. ROM C. RAM D. Hard drive

6. RAM and ROM — Which answer would fit on a poster beside the words "is temporary working memory used while programs run"?

A. ROM B. RAM C. Hard drive D. Storage

7. RAM and ROM — Imagine helping a younger pupil understand something that keeps files for later use. Which term would you teach?

A. Hard drive B. Storage C. RAM D. ROM

8. RAM and ROM — Which concept matches the explanation "is a storage device used in many computers" when you review ram and rom?

A. Storage B. RAM C. ROM D. Hard drive

9. RAM and ROM — For a short demonstration, the teacher needs an example that keeps files for later use. Which choice would you select?

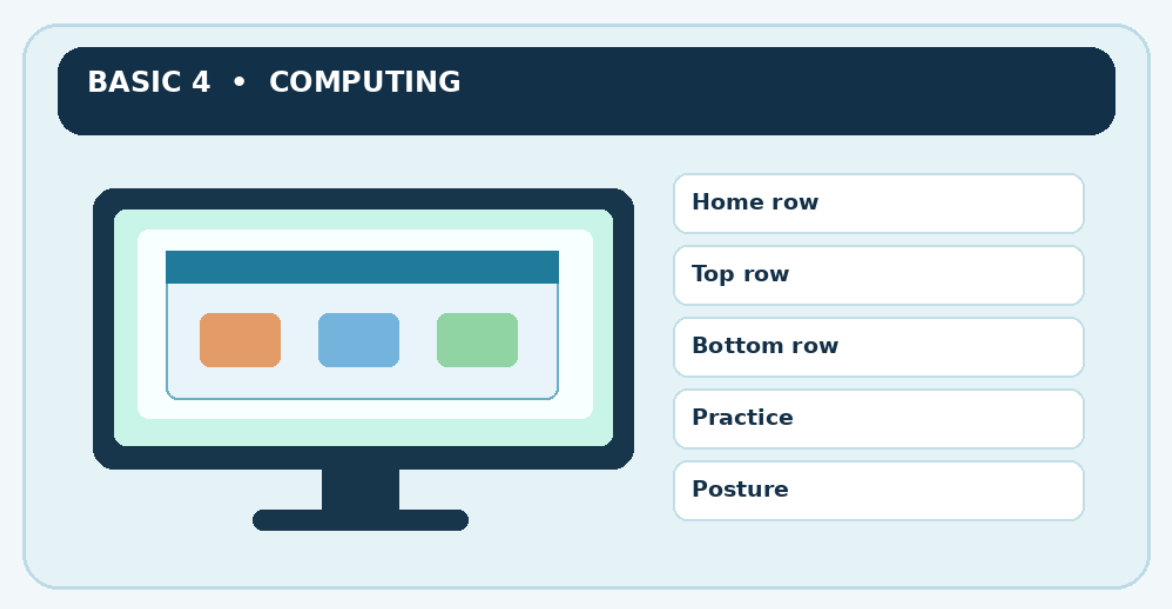
A. Storage B. RAM C. Hard drive D. ROM

10. RAM and ROM — At home, Akosua remembers a class rule or concept that is portable file storage. What was it called?

A. RAM B. Hard drive C. Flash drive D. ROM

Lesson 4: Typing Short Phrases

We will learn to: Identify, explain and apply the important ideas behind typing short phrases.



Five important ideas

Home row: is the resting row for touch typing.

Top row: contains letters Q through P.

Bottom row: contains letters Z through M.

Practice: helps develop accuracy before speed.

Posture: keeps shoulders relaxed while typing.

Our Ghanaian story

At an ICT typing competition, Abena had a question about typing short phrases. The group discovered that home row is the resting row for touch typing, while top row contains letters Q through P. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise home row and top row using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches typing short phrases. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.1.1.4

Lesson 4: Play & Challenge

PLAY — Choose one best answer for each question.

1. Typing Short Phrases — Choose the correct name for something that is the resting row for touch typing.

A. Top row B. Practice C. Home row D. Posture

2. Typing Short Phrases — Your friend says, "This contains letters Q through P." What is the friend describing?

A. Practice B. Bottom row C. Top row D. Home row

3. Typing Short Phrases — Which idea from typing short phrases matches the clue "contains letters Z through M"?

A. Practice B. Posture C. Bottom row D. Home row

4. Typing Short Phrases — The teacher asks for an example that helps develop accuracy before speed. Which answer fits?

A. Posture B. Practice C. Bottom row D. Top row

5. Typing Short Phrases — Can you identify the term described here: keeps shoulders relaxed while typing?

A. Bottom row B. Practice C. Posture D. Home row

CHALLENGE — Think and apply.

1. Typing Short Phrases — During a school ICT lesson, Esi wants to show something that is the resting row for touch typing. Which idea should be labelled?

A. Practice B. Posture C. Top row D. Home row

2. Typing Short Phrases — Which concept matches the explanation "keeps shoulders relaxed while typing" when you review typing short phrases?

A. Bottom row B. Practice C. Posture D. Home row

3. Typing Short Phrases — A group makes a display on typing short phrases. Their first clue is "contains letters Q through P". Which label is correct?

A. Home row B. Bottom row C. Practice D. Top row

4. Typing Short Phrases — For a short demonstration, the teacher needs an example that contains letters Z through M. Which choice would you select?

A. Posture B. Practice C. Home row D. Bottom row

5. Typing Short Phrases — At the community learning centre, someone describes a tool or idea that keeps shoulders relaxed while typing. Which one is it?

A. Practice B. Bottom row C. Posture D. Home row

6. Typing Short Phrases — Which answer would fit on a poster beside the words "is the resting row for touch typing"?

A. Practice B. Top row C. Home row D. Posture

7. Typing Short Phrases — A classmate confuses several ideas about typing short phrases. Which one actually contains letters Q through P?

A. Bottom row B. Practice C. Home row D. Top row

8. Typing Short Phrases — At home, Esi remembers a class rule or concept that helps develop accuracy before speed. What was it called?

A. Top row B. Posture C. Bottom row D. Practice

9. Typing Short Phrases — Your class is solving a practical problem. One useful thing helps develop accuracy before speed. What is its name?

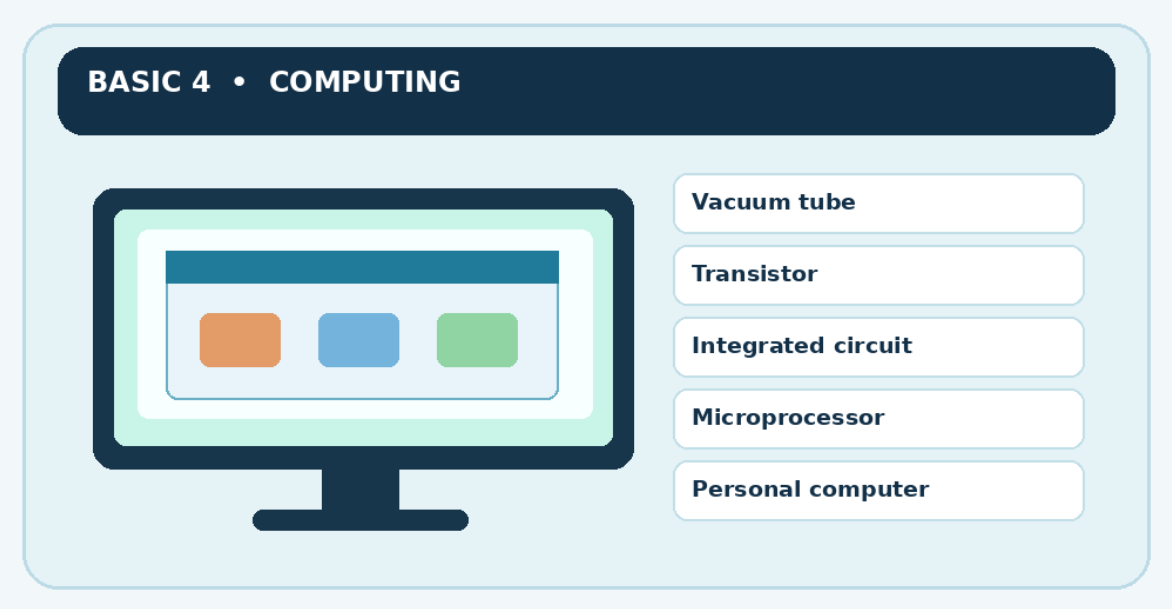
A. Top row B. Posture C. Bottom row D. Practice

10. Typing Short Phrases — Imagine helping a younger pupil understand something that contains letters Z through M. Which term would you teach?

A. Home row B. Posture C. Practice D. Bottom row

Lesson 5: First Generation Computers

We will learn to: Identify, explain and apply the important ideas behind first generation computers.



Five important ideas

Vacuum tube: was used in many first-generation computers.

Transistor: replaced vacuum tubes in many later machines.

Integrated circuit: combined electronic parts on a chip.

Microprocessor: puts processing circuitry onto a chip.

Personal computer: is designed for use by an individual.

Our Ghanaian story

While looking at old and new computer photographs, Kofi had a question about first generation computers. The group discovered that vacuum tube was used in many first-generation computers, while transistor replaced vacuum tubes in many later machines. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of vacuum tube, transistor and integrated circuit. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches first generation computers. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.1.1.5

Lesson 5: Play & Challenge

PLAY — Choose one best answer for each question.

1. First Generation Computers — Choose the correct name for something that was used in many first-generation computers.

A. Integrated circuit B. Personal computer C. Microprocessor D. Vacuum tube

2. First Generation Computers — Your friend says, "This replaced vacuum tubes in many later machines." What is the friend describing?

A. Transistor B. Personal computer C. Microprocessor D. Vacuum tube

3. First Generation Computers — Which idea from first generation computers matches the clue "combined electronic parts on a chip"?

A. Integrated circuit B. Personal computer C. Vacuum tube D. Transistor

4. First Generation Computers — The teacher asks for an example that puts processing circuitry onto a chip. Which answer fits?

A. Personal computer B. Microprocessor C. Vacuum tube D. Transistor

5. First Generation Computers — Can you identify the term described here: is designed for use by an individual?

A. Vacuum tube B. Transistor C. Personal computer D. Integrated circuit

CHALLENGE — Think and apply.

1. First Generation Computers — Imagine helping a younger pupil understand something that combined electronic parts on a chip. Which term would you teach?

A. Integrated circuit B. Vacuum tube C. Personal computer D. Transistor

2. First Generation Computers — Which concept matches the explanation "is designed for use by an individual" when you review first generation computers?

A. Transistor B. Integrated circuit C. Vacuum tube D. Personal computer

3. First Generation Computers — A group makes a display on first generation computers. Their first clue is "replaced vacuum tubes in many later machines". Which label is correct?

A. Transistor B. Vacuum tube C. Personal computer D. Microprocessor

4. First Generation Computers — A classmate confuses several ideas about first generation computers. Which one actually replaced vacuum tubes in many later machines?

A. Vacuum tube B. Transistor C. Personal computer D. Microprocessor

5. First Generation Computers — Your class is solving a practical problem. One useful thing puts processing circuitry onto a chip. What is its name?

A. Personal computer B. Microprocessor C. Transistor D. Vacuum tube

6. First Generation Computers — At home, Ama remembers a class rule or concept that puts processing circuitry onto a chip. What was it called?

A. Personal computer B. Microprocessor C. Vacuum tube D. Transistor

7. First Generation Computers — For a short demonstration, the teacher needs an example that combined electronic parts on a chip. Which choice would you select?

A. Vacuum tube B. Personal computer C. Transistor D. Integrated circuit

8. First Generation Computers — During a school ICT lesson, Ama wants to show something that was used in many first-generation computers. Which idea should be labelled?

A. Personal computer B. Vacuum tube C. Integrated circuit D. Microprocessor

9. First Generation Computers — Which answer would fit on a poster beside the words "was used in many first-generation computers"?

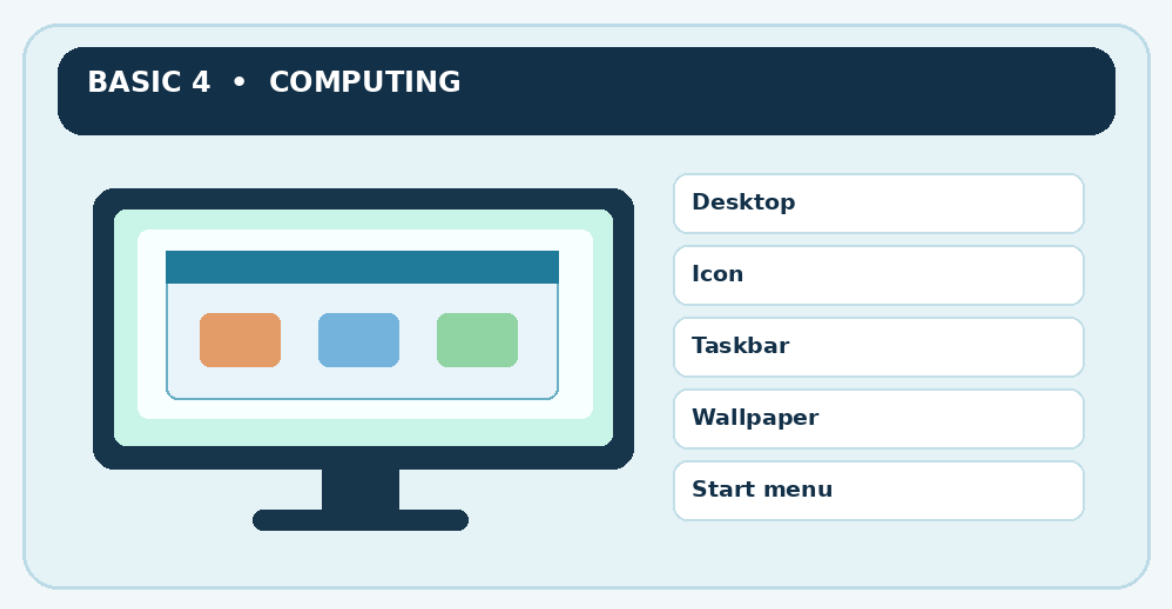
A. Microprocessor B. Integrated circuit C. Vacuum tube D. Personal computer

10. First Generation Computers — At the community learning centre, someone describes a tool or idea that is designed for use by an individual. Which one is it?

A. Personal computer B. Vacuum tube C. Transistor D. Integrated circuit

Lesson 6: Desktop Features

We will learn to: Identify, explain and apply the important ideas behind desktop features.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Ama had a question about desktop features. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches desktop features. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.1

Lesson 6: Play & Challenge

PLAY — Choose one best answer for each question.

1. Desktop Features — Choose the correct name for something that is a main computer work area.

A. Icon B. Taskbar C. Start menu D. Desktop

2. Desktop Features — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Start menu B. Wallpaper C. Taskbar D. Icon

3. Desktop Features — Which idea from desktop features matches the clue "helps find open apps and tools"?

A. Taskbar B. Start menu C. Desktop D. Wallpaper

4. Desktop Features — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Wallpaper B. Icon C. Start menu D. Taskbar

5. Desktop Features — Can you identify the term described here: offers a way to open programs?

A. Icon B. Desktop C. Wallpaper D. Start menu

CHALLENGE — Think and apply.

1. Desktop Features — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Start menu B. Taskbar C. Desktop D. Icon

2. Desktop Features — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

A. Wallpaper B. Start menu C. Icon D. Taskbar

3. Desktop Features — During a school ICT lesson, Yaw wants to show something that is a main computer work area. Which idea should be labelled?

A. Desktop B. Taskbar C. Icon D. Start menu

4. Desktop Features — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Start menu B. Taskbar C. Desktop D. Wallpaper

5. Desktop Features — At home, Yaw remembers a class rule or concept that is the desktop background picture. What was it called?

A. Wallpaper B. Icon C. Start menu D. Taskbar

6. Desktop Features — Which concept matches the explanation "offers a way to open programs" when you review desktop features?

A. Desktop B. Wallpaper C. Start menu D. Icon

7. Desktop Features — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Icon B. Start menu C. Desktop D. Wallpaper

8. Desktop Features — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Taskbar B. Desktop C. Wallpaper D. Start menu

9. Desktop Features — A classmate confuses several ideas about desktop features. Which one actually is a small picture representing an item?

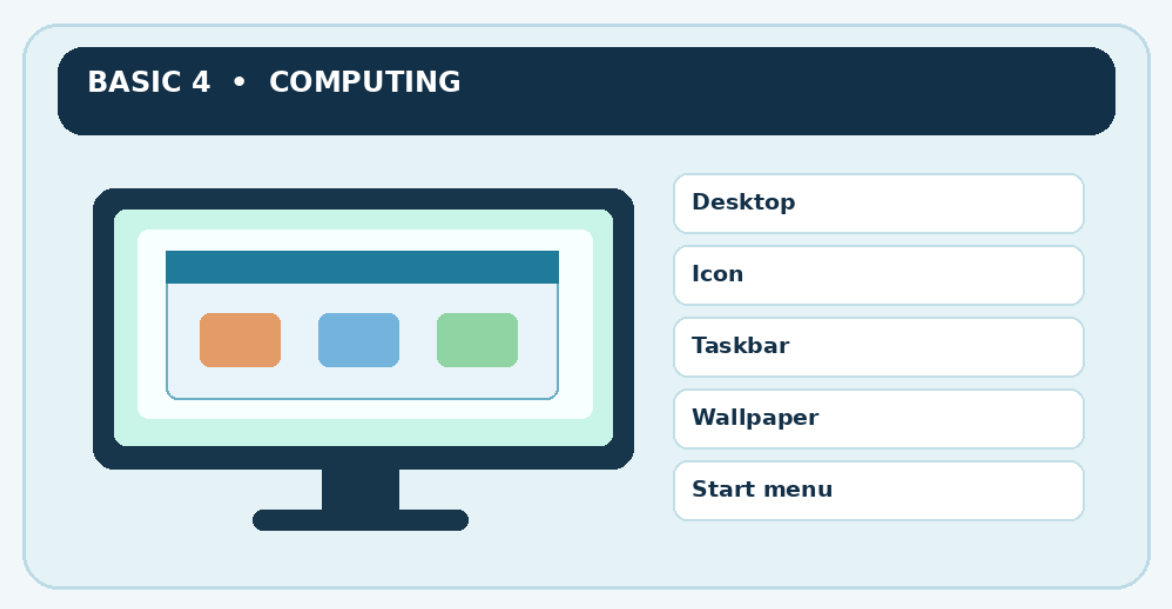
A. Start menu B. Wallpaper C. Taskbar D. Icon

10. Desktop Features — A group makes a display on desktop features. Their first clue is "is a small picture representing an item". Which label is correct?

A. Icon B. Start menu C. Taskbar D. Wallpaper

Lesson 7: Personalise the Desktop

We will learn to: Identify, explain and apply the important ideas behind personalise the desktop.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Kojo had a question about personalise the desktop. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches personalise the desktop. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.2

Lesson 7: Play & Challenge

PLAY — Choose one best answer for each question.

1. Personalise the Desktop — Choose the correct name for something that is a main computer work area.

A. Wallpaper B. Taskbar C. Icon D. Desktop

2. Personalise the Desktop — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Icon B. Start menu C. Taskbar D. Desktop

3. Personalise the Desktop — Which idea from personalise the desktop matches the clue "helps find open apps and tools"?

A. Start menu B. Taskbar C. Icon D. Desktop

4. Personalise the Desktop — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Start menu B. Wallpaper C. Taskbar D. Icon

5. Personalise the Desktop — Can you identify the term described here: offers a way to open programs?

A. Icon B. Start menu C. Taskbar D. Desktop

CHALLENGE — Think and apply.

1. Personalise the Desktop — At home, Kofi remembers a class rule or concept that is the desktop background picture. What was it called?

A. Start menu B. Wallpaper C. Icon D. Taskbar

2. Personalise the Desktop — A classmate confuses several ideas about personalise the desktop. Which one actually is a small picture representing an item?

A. Desktop B. Icon C. Start menu D. Taskbar

3. Personalise the Desktop — A group makes a display on personalise the desktop. Their first clue is "is a small picture representing an item". Which label is correct?

A. Start menu B. Icon C. Desktop D. Taskbar

4. Personalise the Desktop — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Wallpaper B. Desktop C. Taskbar D. Icon

5. Personalise the Desktop — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

A. Start menu B. Taskbar C. Wallpaper D. Icon

6. Personalise the Desktop — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Start menu B. Taskbar C. Icon D. Desktop

7. Personalise the Desktop — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Desktop B. Start menu C. Taskbar D. Icon

8. Personalise the Desktop — Which concept matches the explanation "offers a way to open programs" when you review personalise the desktop?

A. Icon B. Taskbar C. Start menu D. Desktop

9. Personalise the Desktop — During a school ICT lesson, Kofi wants to show something that is a main computer work area. Which idea should be labelled?

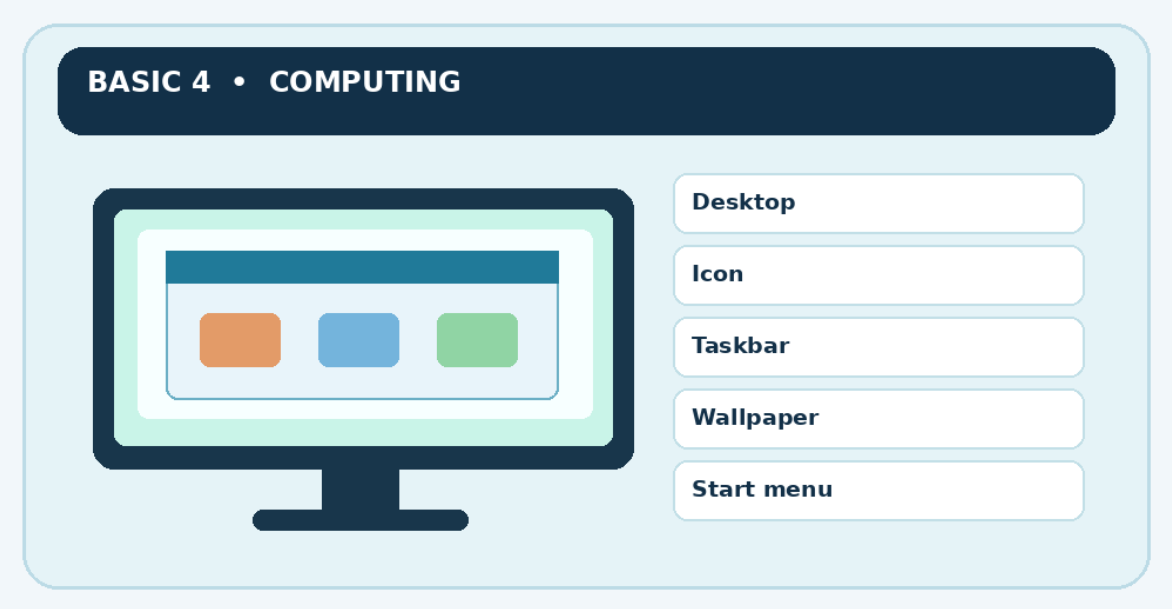
A. Taskbar B. Wallpaper C. Desktop D. Icon

10. Personalise the Desktop — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Start menu B. Icon C. Desktop D. Taskbar

Lesson 8: Taskbar and Start Menu

We will learn to: Identify, explain and apply the important ideas behind taskbar and start menu.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Esi had a question about taskbar and start menu. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches taskbar and start menu. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.3

Lesson 8: Play & Challenge

PLAY — Choose one best answer for each question.

1. Taskbar and Start Menu — Choose the correct name for something that is a main computer work area.

A. Start menu B. Desktop C. Icon D. Wallpaper

2. Taskbar and Start Menu — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Wallpaper B. Taskbar C. Desktop D. Icon

3. Taskbar and Start Menu — Which idea from taskbar and start menu matches the clue "helps find open apps and tools"?

A. Wallpaper B. Desktop C. Icon D. Taskbar

4. Taskbar and Start Menu — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Wallpaper B. Taskbar C. Icon D. Start menu

5. Taskbar and Start Menu — Can you identify the term described here: offers a way to open programs?

A. Icon B. Start menu C. Wallpaper D. Taskbar

CHALLENGE — Think and apply.

1. Taskbar and Start Menu — During a school ICT lesson, Akosua wants to show something that is a main computer work area. Which idea should be labelled?

A. Wallpaper B. Start menu C. Desktop D. Icon

2. Taskbar and Start Menu — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Wallpaper B. Desktop C. Taskbar D. Icon

3. Taskbar and Start Menu — A classmate confuses several ideas about taskbar and start menu. Which one actually is a small picture representing an item?

A. Wallpaper B. Desktop C. Taskbar D. Icon

4. Taskbar and Start Menu — At home, Akosua remembers a class rule or concept that is the desktop background picture. What was it called?

A. Wallpaper B. Start menu C. Taskbar D. Icon

5. Taskbar and Start Menu — Which concept matches the explanation "offers a way to open programs" when you review taskbar and start menu?

A. Taskbar B. Wallpaper C. Icon D. Start menu

6. Taskbar and Start Menu — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Taskbar B. Icon C. Start menu D. Wallpaper

7. Taskbar and Start Menu — A group makes a display on taskbar and start menu. Their first clue is "is a small picture representing an item". Which label is correct?

A. Wallpaper B. Desktop C. Icon D. Taskbar

8. Taskbar and Start Menu — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Start menu B. Icon C. Desktop D. Wallpaper

9. Taskbar and Start Menu — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

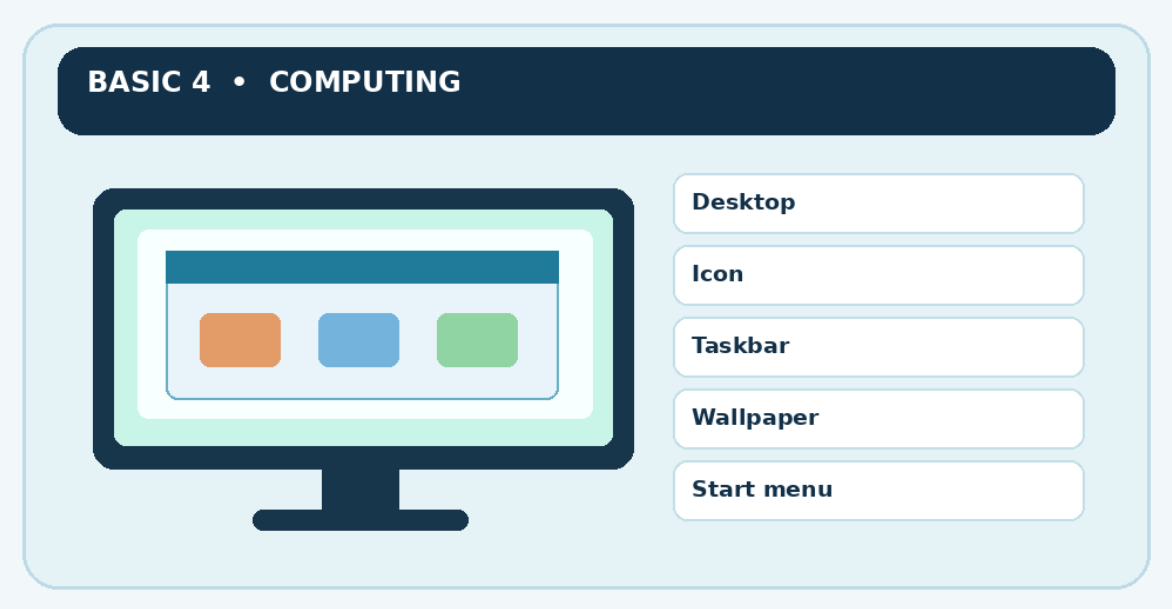
A. Wallpaper B. Icon C. Start menu D. Taskbar

10. Taskbar and Start Menu — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Icon B. Taskbar C. Wallpaper D. Desktop

Lesson 9: Add and Remove Desktop Icons

We will learn to: Identify, explain and apply the important ideas behind add and remove desktop icons.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Yaw had a question about add and remove desktop icons. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches add and remove desktop icons. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.4

Lesson 9: Play & Challenge

PLAY — Choose one best answer for each question.

1. Add and Remove Desktop Icons — Choose the correct name for something that is a main computer work area.

A. Desktop B. Wallpaper C. Start menu D. Taskbar

2. Add and Remove Desktop Icons — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Taskbar B. Desktop C. Wallpaper D. Icon

3. Add and Remove Desktop Icons — Which idea from add and remove desktop icons matches the clue "helps find open apps and tools"?

A. Taskbar B. Wallpaper C. Start menu D. Icon

4. Add and Remove Desktop Icons — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Taskbar B. Wallpaper C. Icon D. Desktop

5. Add and Remove Desktop Icons — Can you identify the term described here: offers a way to open programs?

A. Wallpaper B. Taskbar C. Icon D. Start menu

CHALLENGE — Think and apply.

1. Add and Remove Desktop Icons — Which concept matches the explanation "offers a way to open programs" when you review add and remove desktop icons?

A. Wallpaper B. Icon C. Taskbar D. Start menu

2. Add and Remove Desktop Icons — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Icon B. Start menu C. Taskbar D. Wallpaper

3. Add and Remove Desktop Icons — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Taskbar B. Desktop C. Wallpaper D. Start menu

4. Add and Remove Desktop Icons — During a school ICT lesson, Esi wants to show something that is a main computer work area. Which idea should be labelled?

A. Wallpaper B. Desktop C. Start menu D. Taskbar

5. Add and Remove Desktop Icons — A classmate confuses several ideas about add and remove desktop icons. Which one actually is a small picture representing an item?

A. Wallpaper B. Desktop C. Icon D. Taskbar

6. Add and Remove Desktop Icons — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

A. Taskbar B. Icon C. Desktop D. Wallpaper

7. Add and Remove Desktop Icons — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Taskbar B. Icon C. Wallpaper D. Start menu

8. Add and Remove Desktop Icons — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Taskbar B. Start menu C. Icon D. Wallpaper

9. Add and Remove Desktop Icons — At home, Esi remembers a class rule or concept that is the desktop background picture. What was it called?

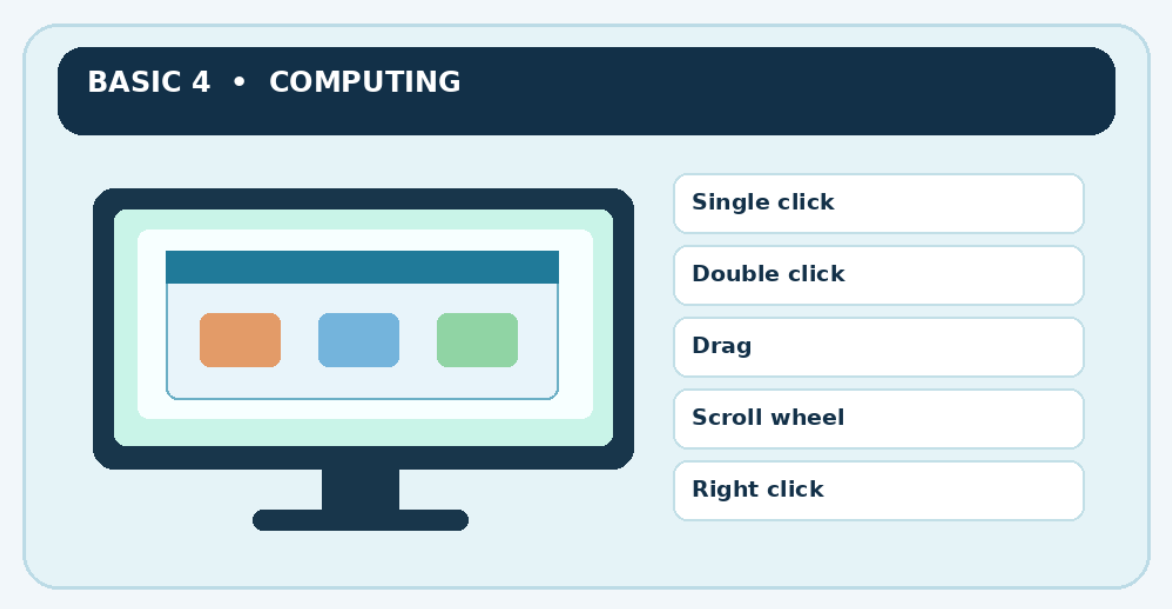
A. Taskbar B. Desktop C. Icon D. Wallpaper

10. Add and Remove Desktop Icons — A group makes a display on add and remove desktop icons. Their first clue is "is a small picture representing an item". Which label is correct?

A. Desktop B. Wallpaper C. Taskbar D. Icon

Lesson 10: Moving and Selecting Icons

We will learn to: Identify, explain and apply the important ideas behind moving and selecting icons.



Five important ideas

Single click: selects an item.

Double click: opens many files or apps.

Drag: holds the button while moving an item.

Scroll wheel: moves through a page.

Right click: opens a context menu in many systems.

Our Ghanaian story

At a shared classroom computer, Abena had a question about moving and selecting icons. The group discovered that single click selects an item, while double click opens many files or apps. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise single click and double click using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches moving and selecting icons. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.5

Lesson 10: Play & Challenge

PLAY — Choose one best answer for each question.

1. Moving and Selecting Icons — Choose the correct name for something that selects an item.

A. Single click B. Drag C. Scroll wheel D. Double click

2. Moving and Selecting Icons — Your friend says, "This opens many files or apps." What is the friend describing?

A. Scroll wheel B. Double click C. Drag D. Single click

3. Moving and Selecting Icons — Which idea from moving and selecting icons matches the clue "holds the button while moving an item"?

A. Single click B. Scroll wheel C. Drag D. Right click

4. Moving and Selecting Icons — The teacher asks for an example that moves through a page. Which answer fits?

A. Right click B. Drag C. Scroll wheel D. Single click

5. Moving and Selecting Icons — Can you identify the term described here: opens a context menu in many systems?

A. Right click B. Drag C. Scroll wheel D. Double click

CHALLENGE — Think and apply.

1. Moving and Selecting Icons — For a short demonstration, the teacher needs an example that holds the button while moving an item. Which choice would you select?

A. Right click B. Scroll wheel C. Drag D. Single click

2. Moving and Selecting Icons — During a school ICT lesson, Ama wants to show something that selects an item. Which idea should be labelled?

A. Drag B. Single click C. Double click D. Scroll wheel

3. Moving and Selecting Icons — A classmate confuses several ideas about moving and selecting icons. Which one actually opens many files or apps?

A. Drag B. Double click C. Scroll wheel D. Single click

4. Moving and Selecting Icons — Which answer would fit on a poster beside the words "selects an item"?

A. Double click B. Single click C. Drag D. Scroll wheel

5. Moving and Selecting Icons — Which concept matches the explanation "opens a context menu in many systems" when you review moving and selecting icons?

A. Double click B. Drag C. Scroll wheel D. Right click

6. Moving and Selecting Icons — Imagine helping a younger pupil understand something that holds the button while moving an item. Which term would you teach?

A. Scroll wheel B. Right click C. Single click D. Drag

7. Moving and Selecting Icons — At home, Ama remembers a class rule or concept that moves through a page. What was it called?

A. Drag B. Right click C. Scroll wheel D. Single click

8. Moving and Selecting Icons — Your class is solving a practical problem. One useful thing moves through a page. What is its name?

A. Single click B. Drag C. Right click D. Scroll wheel

9. Moving and Selecting Icons — A group makes a display on moving and selecting icons. Their first clue is "opens many files or apps". Which label is correct?

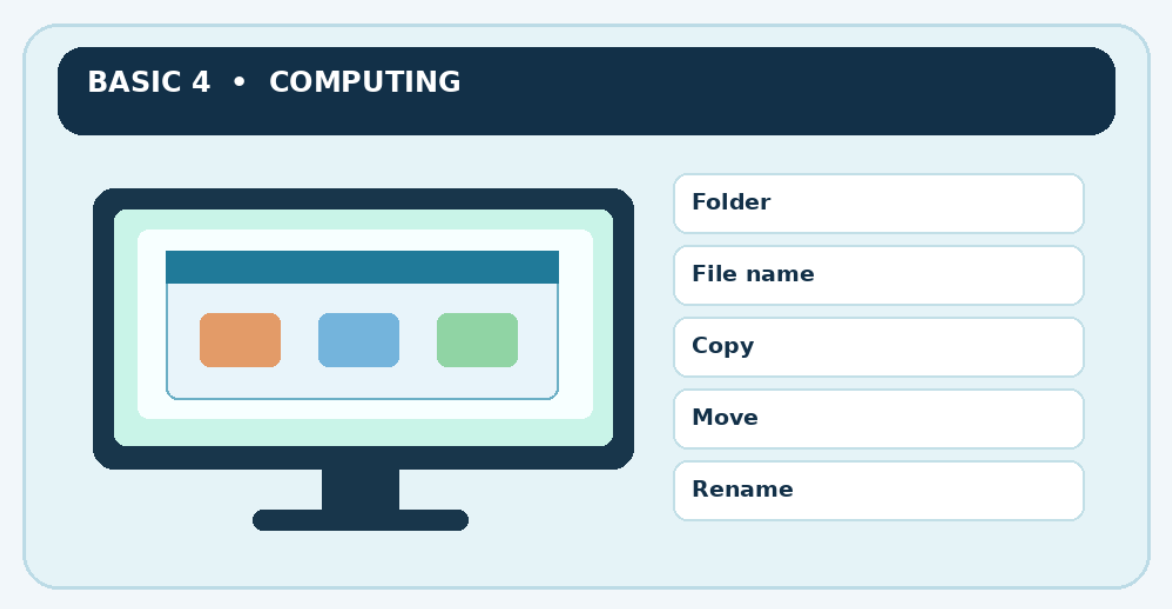
A. Double click B. Drag C. Single click D. Scroll wheel

10. Moving and Selecting Icons — At the community learning centre, someone describes a tool or idea that opens a context menu in many systems. Which one is it?

A. Drag B. Right click C. Scroll wheel D. Double click

Lesson 11: Creating and Renaming Folders

We will learn to: Identify, explain and apply the important ideas behind creating and renaming folders.



Five important ideas

Folder: organises related digital files.

File name: helps identify a saved item.

Copy: makes a second version of an item.

Move: changes where an item is stored.

Rename: changes the label without changing the file content.

Our Ghanaian story

While organising class projects, Kofi had a question about creating and renaming folders. The group discovered that folder organises related digital files, while file name helps identify a saved item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise folder and file name using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches creating and renaming folders. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.2.1.6

Lesson 11: Play & Challenge

PLAY — Choose one best answer for each question.

1. Creating and Renaming Folders — Choose the correct name for something that organises related digital files.

A. Folder B. File name C. Rename D. Move

2. Creating and Renaming Folders — Your friend says, "This helps identify a saved item." What is the friend describing?

A. Rename B. Folder C. Copy D. File name

3. Creating and Renaming Folders — Which idea from creating and renaming folders matches the clue "makes a second version of an item"?

A. Copy B. Rename C. Folder D. File name

4. Creating and Renaming Folders — The teacher asks for an example that changes where an item is stored. Which answer fits?

A. Folder B. Move C. Rename D. File name

5. Creating and Renaming Folders — Can you identify the term described here: changes the label without changing the file content?

A. Rename B. Move C. Folder D. Copy

CHALLENGE — Think and apply.

1. Creating and Renaming Folders — Which concept matches the explanation "changes the label without changing the file content" when you review creating and renaming folders?

A. Copy B. Move C. Folder D. Rename

2. Creating and Renaming Folders — During a school ICT lesson, Yaw wants to show something that organises related digital files. Which idea should be labelled?

A. Folder B. Rename C. Move D. File name

3. Creating and Renaming Folders — At home, Yaw remembers a class rule or concept that changes where an item is stored. What was it called?

A. Folder B. Move C. Rename D. File name

4. Creating and Renaming Folders — A classmate confuses several ideas about creating and renaming folders. Which one actually helps identify a saved item?

A. File name B. Copy C. Folder D. Rename

5. Creating and Renaming Folders — At the community learning centre, someone describes a tool or idea that changes the label without changing the file content. Which one is it?

A. Folder B. Rename C. Copy D. Move

6. Creating and Renaming Folders — For a short demonstration, the teacher needs an example that makes a second version of an item. Which choice would you select?

A. Rename B. Folder C. Copy D. File name

7. Creating and Renaming Folders — A group makes a display on creating and renaming folders. Their first clue is "helps identify a saved item". Which label is correct?

A. Copy B. Folder C. File name D. Rename

8. Creating and Renaming Folders — Imagine helping a younger pupil understand something that makes a second version of an item. Which term would you teach?

A. Folder B. Rename C. Copy D. File name

9. Creating and Renaming Folders — Which answer would fit on a poster beside the words "organises related digital files"?

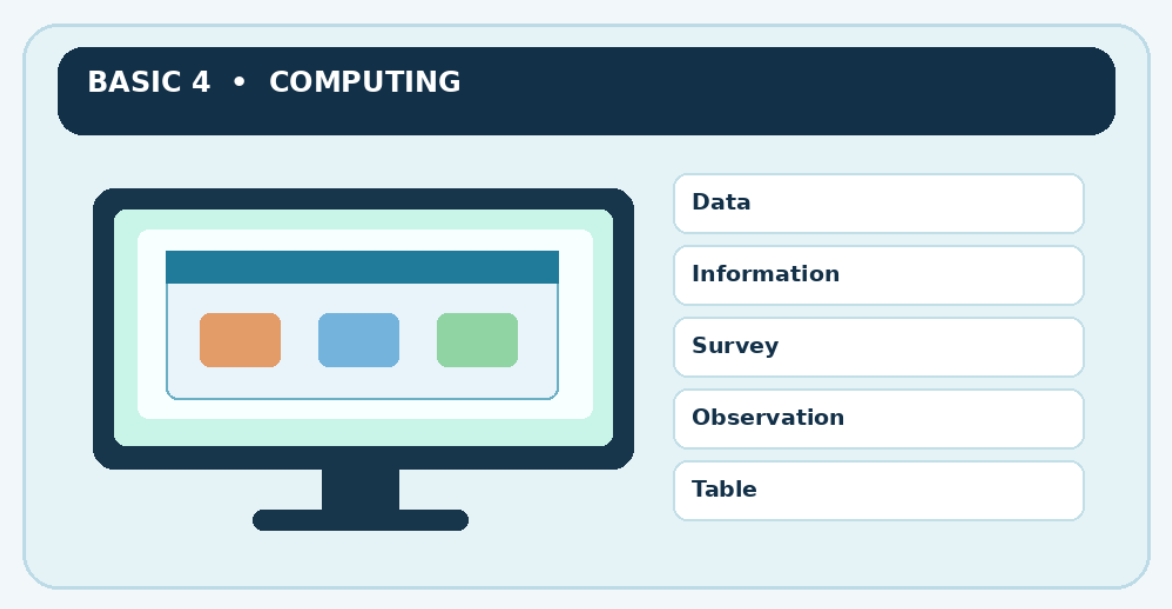
A. Folder B. File name C. Rename D. Move

10. Creating and Renaming Folders — Your class is solving a practical problem. One useful thing changes where an item is stored. What is its name?

A. File name B. Folder C. Move D. Rename

Lesson 12: Identify Data in Everyday Reports

We will learn to: Identify, explain and apply the important ideas behind identify data in everyday reports.



Five important ideas

Data: raw facts gathered for a purpose.

Information: data organised to make sense.

Survey: asks people selected questions.

Observation: records what someone notices.

Table: arranges facts into rows and columns.

Our Ghanaian story

During a survey of the school, Ama had a question about identify data in everyday reports. The group discovered that data raw facts gathered for a purpose, while information data organised to make sense. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Data, Information and Survey; explain what each means.

Create and share

Make a colourful, labelled poster that teaches identify data in everyday reports. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.1.1

Lesson 12: Play & Challenge

PLAY — Choose one best answer for each question.

1. Identify Data in Everyday Reports — Choose the correct name for something that raw facts gathered for a purpose.

A. Table B. Survey C. Data D. Observation

2. Identify Data in Everyday Reports — Your friend says, "This data organised to make sense." What is the friend describing?

A. Information B. Data C. Observation D. Table

3. Identify Data in Everyday Reports — Which idea from identify data in everyday reports matches the clue "asks people selected questions"?

A. Survey B. Data C. Observation D. Table

4. Identify Data in Everyday Reports — The teacher asks for an example that records what someone notices. Which answer fits?

A. Survey B. Table C. Observation D. Information

5. Identify Data in Everyday Reports — Can you identify the term described here: arranges facts into rows and columns?

A. Survey B. Table C. Information D. Data

CHALLENGE — Think and apply.

1. Identify Data in Everyday Reports — At home, Kofi remembers a class rule or concept that records what someone notices. What was it called?

A. Observation B. Table C. Survey D. Information

2. Identify Data in Everyday Reports — At the community learning centre, someone describes a tool or idea that arranges facts into rows and columns. Which one is it?

A. Survey B. Data C. Table D. Information

3. Identify Data in Everyday Reports — Your class is solving a practical problem. One useful thing records what someone notices. What is its name?

A. Information B. Observation C. Table D. Survey

4. Identify Data in Everyday Reports — Which concept matches the explanation "arranges facts into rows and columns" when you review identify data in everyday reports?

A. Table B. Data C. Survey D. Information

5. Identify Data in Everyday Reports — Imagine helping a younger pupil understand something that asks people selected questions. Which term would you teach?

A. Data B. Table C. Survey D. Observation

6. Identify Data in Everyday Reports — For a short demonstration, the teacher needs an example that asks people selected questions. Which choice would you select?

A. Data B. Observation C. Table D. Survey

7. Identify Data in Everyday Reports — During a school ICT lesson, Kofi wants to show something that raw facts gathered for a purpose. Which idea should be labelled?

A. Data B. Table C. Observation D. Survey

8. Identify Data in Everyday Reports — A classmate confuses several ideas about identify data in everyday reports. Which one actually data organised to make sense?

A. Table B. Data C. Information D. Observation

9. Identify Data in Everyday Reports — A group makes a display on identify data in everyday reports. Their first clue is "data organised to make sense". Which label is correct?

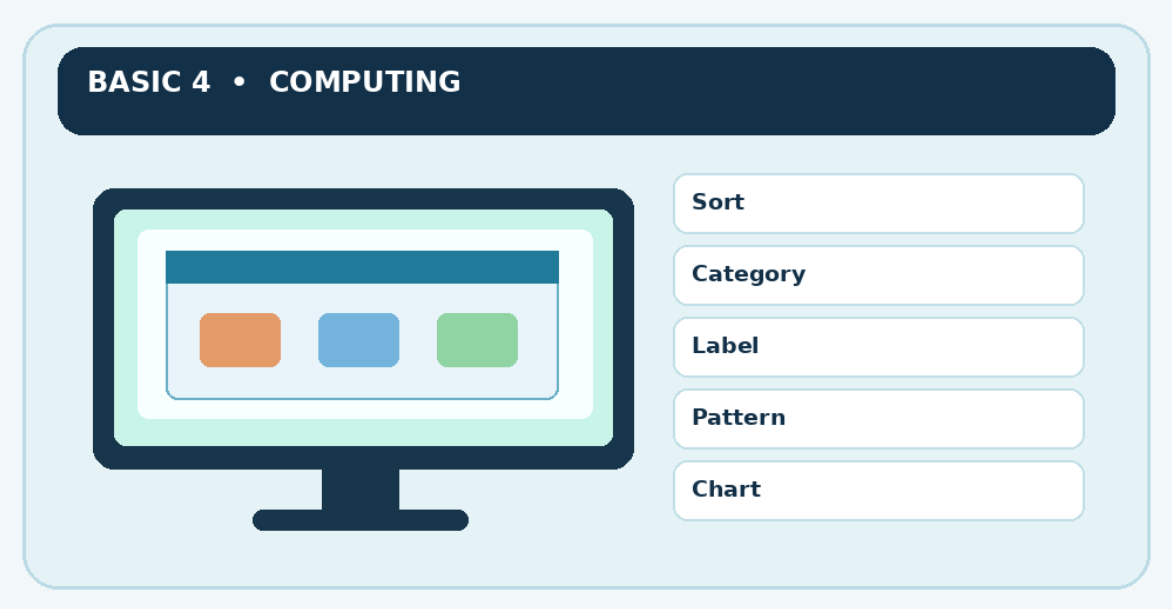
A. Observation B. Information C. Table D. Data

10. Identify Data in Everyday Reports — Which answer would fit on a poster beside the words "raw facts gathered for a purpose"?

A. Table B. Observation C. Survey D. Data

Lesson 13: Classify Data into Information

We will learn to: Identify, explain and apply the important ideas behind classify data into information.



Five important ideas

Sort: arranges data into an order.

Category: groups items sharing a feature.

Label: gives a clear name to a group.

Pattern: is a regular feature seen in data.

Chart: shows data visually.

Our Ghanaian story

When sorting market-day information, Kojo had a question about classify data into information. The group discovered that sort arranges data into an order, while category groups items sharing a feature. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Sort, Category and Label; explain what each means.

Create and share

Make a colourful, labelled poster that teaches classify data into information. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.1.2

Lesson 13: Play & Challenge

PLAY — Choose one best answer for each question.

1. Classify Data into Information — Choose the correct name for something that arranges data into an order.

A. Chart B. Sort C. Pattern D. Category

2. Classify Data into Information — Your friend says, "This groups items sharing a feature." What is the friend describing?

A. Chart B. Category C. Sort D. Label

3. Classify Data into Information — Which idea from classify data into information matches the clue "gives a clear name to a group"?

A. Chart B. Label C. Category D. Sort

4. Classify Data into Information — The teacher asks for an example that is a regular feature seen in data. Which answer fits?

A. Sort B. Pattern C. Chart D. Category

5. Classify Data into Information — Can you identify the term described here: shows data visually?

A. Category B. Chart C. Sort D. Label

CHALLENGE — Think and apply.

1. Classify Data into Information — Your class is solving a practical problem. One useful thing is a regular feature seen in data. What is its name?

A. Sort B. Pattern C. Category D. Chart

2. Classify Data into Information — At home, Akosua remembers a class rule or concept that is a regular feature seen in data. What was it called?

A. Chart B. Category C. Pattern D. Sort

3. Classify Data into Information — Imagine helping a younger pupil understand something that gives a clear name to a group. Which term would you teach?

A. Sort B. Label C. Chart D. Category

4. Classify Data into Information — For a short demonstration, the teacher needs an example that gives a clear name to a group. Which choice would you select?

A. Sort B. Chart C. Category D. Label

5. Classify Data into Information — During a school ICT lesson, Akosua wants to show something that arranges data into an order. Which idea should be labelled?

A. Pattern B. Category C. Sort D. Chart

6. Classify Data into Information — A classmate confuses several ideas about classify data into information. Which one actually groups items sharing a feature?

A. Label B. Category C. Chart D. Sort

7. Classify Data into Information — Which answer would fit on a poster beside the words "arranges data into an order"?

A. Pattern B. Category C. Sort D. Chart

8. Classify Data into Information — Which concept matches the explanation "shows data visually" when you review classify data into information?

A. Sort B. Label C. Chart D. Category

9. Classify Data into Information — At the community learning centre, someone describes a tool or idea that shows data visually. Which one is it?

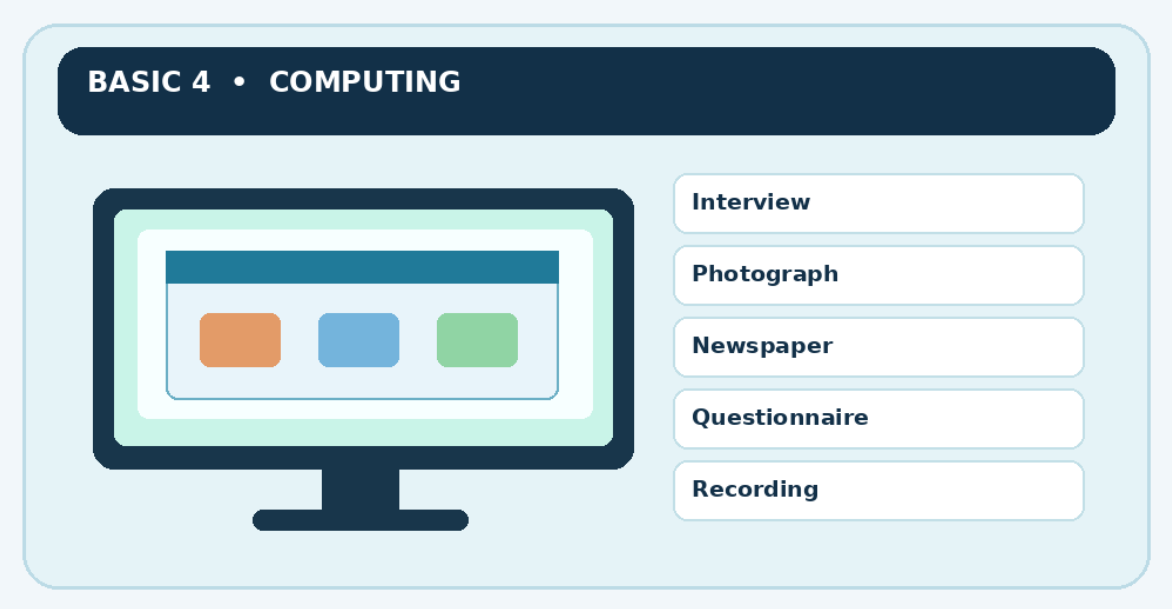
A. Label B. Category C. Sort D. Chart

10. Classify Data into Information — A group makes a display on classify data into information. Their first clue is "groups items sharing a feature". Which label is correct?

A. Label B. Sort C. Chart D. Category

Lesson 14: Recognising Information Sources

We will learn to: Identify, explain and apply the important ideas behind recognising information sources.



Five important ideas

Interview: collects answers by talking to people.

Photograph: records a visual moment.

Newspaper: reports news and events.

Questionnaire: collects written responses.

Recording: stores captured sound or video.

Our Ghanaian story

When preparing a report about the neighbourhood, Esi had a question about recognising information sources. The group discovered that interview collects answers by talking to people, while photograph records a visual moment. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Interview, Photograph and Newspaper; explain what each means.

Create and share

Make a colourful, labelled poster that teaches recognising information sources. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.1.3

Lesson 14: Play & Challenge

PLAY — Choose one best answer for each question.

1. Recognising Information Sources — Choose the correct name for something that collects answers by talking to people.

A. Questionnaire B. Recording C. Interview D. Newspaper

2. Recognising Information Sources — Your friend says, "This records a visual moment." What is the friend describing?

A. Interview B. Newspaper C. Photograph D. Recording

3. Recognising Information Sources — Which idea from recognising information sources matches the clue "reports news and events"?

A. Newspaper B. Questionnaire C. Photograph D. Recording

4. Recognising Information Sources — The teacher asks for an example that collects written responses. Which answer fits?

A. Newspaper B. Interview C. Questionnaire D. Recording

5. Recognising Information Sources — Can you identify the term described here: stores captured sound or video?

A. Newspaper B. Interview C. Recording D. Questionnaire

CHALLENGE — Think and apply.

1. Recognising Information Sources — At home, Esi remembers a class rule or concept that collects written responses. What was it called?

A. Interview B. Questionnaire C. Recording D. Newspaper

2. Recognising Information Sources — A classmate confuses several ideas about recognising information sources. Which one actually records a visual moment?

A. Interview B. Newspaper C. Recording D. Photograph

3. Recognising Information Sources — Which concept matches the explanation "stores captured sound or video" when you review recognising information sources?

A. Recording B. Interview C. Questionnaire D. Newspaper

4. Recognising Information Sources — A group makes a display on recognising information sources. Their first clue is "records a visual moment". Which label is correct?

A. Newspaper B. Photograph C. Recording D. Interview

5. Recognising Information Sources — Which answer would fit on a poster beside the words "collects answers by talking to people"?

A. Newspaper B. Interview C. Questionnaire D. Recording

6. Recognising Information Sources — Your class is solving a practical problem. One useful thing collects written responses. What is its name?

A. Interview B. Questionnaire C. Recording D. Newspaper

7. Recognising Information Sources — For a short demonstration, the teacher needs an example that reports news and events. Which choice would you select?

A. Photograph B. Newspaper C. Questionnaire D. Recording

8. Recognising Information Sources — Imagine helping a younger pupil understand something that reports news and events. Which term would you teach?

A. Questionnaire B. Recording C. Newspaper D. Photograph

9. Recognising Information Sources — During a school ICT lesson, Esi wants to show something that collects answers by talking to people. Which idea should be labelled?

A. Newspaper B. Interview C. Questionnaire D. Recording

10. Recognising Information Sources — At the community learning centre, someone describes a tool or idea that stores captured sound or video. Which one is it?

A. Recording B. Newspaper C. Questionnaire D. Interview

Lesson 15: Sharing Information With Devices

We will learn to: Identify, explain and apply the important ideas behind sharing information with devices.



Five important ideas

Mobile phone: allows calls and messages.

Radio: broadcasts sound to listeners.

Television: shares moving pictures and sound.

Email: sends digital written messages.

Video call: lets distant people see and hear one another.

Our Ghanaian story

When contacting relatives in another region, Yaw had a question about sharing information with devices. The group discovered that mobile phone allows calls and messages, while radio broadcasts sound to listeners. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of mobile phone, radio and television. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches sharing information with devices. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.1.4

Lesson 15: Play & Challenge

PLAY — Choose one best answer for each question.

1. Sharing Information With Devices — Choose the correct name for something that allows calls and messages.

A. Television B. Mobile phone C. Radio D. Video call

2. Sharing Information With Devices — Your friend says, "This broadcasts sound to listeners." What is the friend describing?

A. Television B. Video call C. Radio D. Email

3. Sharing Information With Devices — Which idea from sharing information with devices matches the clue "shares moving pictures and sound"?

A. Mobile phone B. Video call C. Email D. Television

4. Sharing Information With Devices — The teacher asks for an example that sends digital written messages. Which answer fits?

A. Radio B. Mobile phone C. Video call D. Email

5. Sharing Information With Devices — Can you identify the term described here: lets distant people see and hear one another?

A. Email B. Radio C. Video call D. Mobile phone

CHALLENGE — Think and apply.

1. Sharing Information With Devices — Which answer would fit on a poster beside the words "allows calls and messages"?

A. Television B. Video call C. Radio D. Mobile phone

2. Sharing Information With Devices — At home, Ama remembers a class rule or concept that sends digital written messages. What was it called?

A. Radio B. Video call C. Mobile phone D. Email

3. Sharing Information With Devices — During a school ICT lesson, Ama wants to show something that allows calls and messages. Which idea should be labelled?

A. Mobile phone B. Radio C. Video call D. Television

4. Sharing Information With Devices — Imagine helping a younger pupil understand something that shares moving pictures and sound. Which term would you teach?

A. Email B. Mobile phone C. Video call D. Television

5. Sharing Information With Devices — For a short demonstration, the teacher needs an example that shares moving pictures and sound. Which choice would you select?

A. Television B. Video call C. Email D. Mobile phone

6. Sharing Information With Devices — Which concept matches the explanation "lets distant people see and hear one another" when you review sharing information with devices?

A. Radio B. Video call C. Email D. Mobile phone

7. Sharing Information With Devices — A classmate confuses several ideas about sharing information with devices. Which one actually broadcasts sound to listeners?

A. Radio B. Email C. Television D. Video call

8. Sharing Information With Devices — A group makes a display on sharing information with devices. Their first clue is "broadcasts sound to listeners". Which label is correct?

A. Radio B. Email C. Video call D. Television

9. Sharing Information With Devices — At the community learning centre, someone describes a tool or idea that lets distant people see and hear one another. Which one is it?

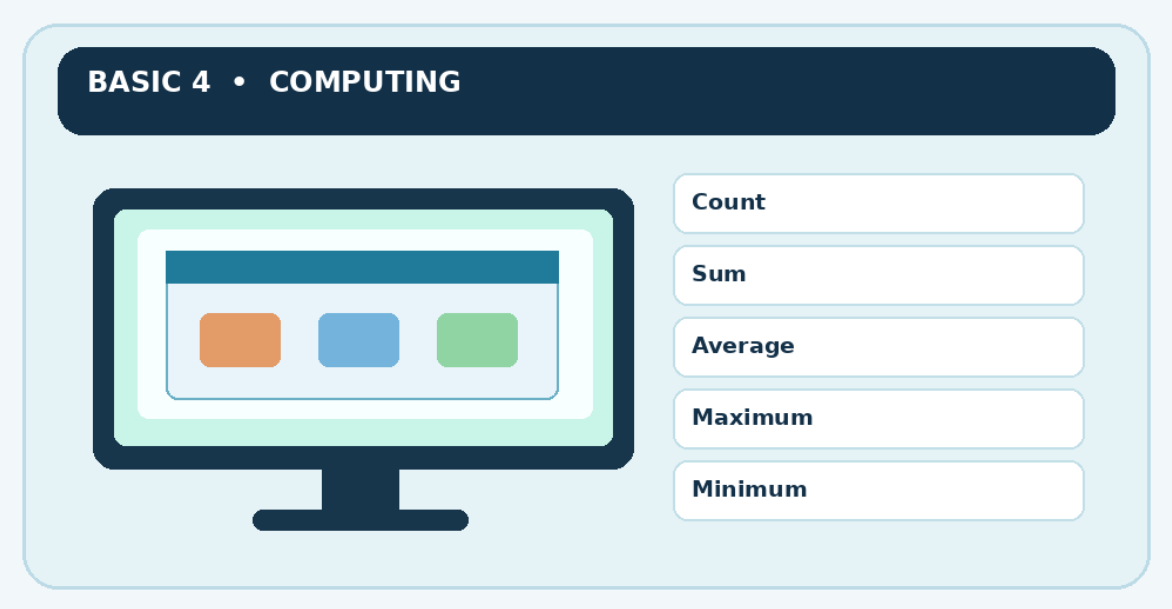
A. Radio B. Video call C. Email D. Mobile phone

10. Sharing Information With Devices — Your class is solving a practical problem. One useful thing sends digital written messages. What is its name?

A. Email B. Radio C. Video call D. Mobile phone

Lesson 16: Calculating With Simple Data

We will learn to: Identify, explain and apply the important ideas behind calculating with simple data.



Five important ideas

Count: tells how many items are present.

Sum: adds numbers together.

Average: shares a total evenly across the number of values.

Maximum: is the largest value.

Minimum: is the smallest value.

Our Ghanaian story

While calculating scores for sports day, Abena had a question about calculating with simple data. The group discovered that count tells how many items are present, while sum adds numbers together. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Count, Sum and Average; explain what each means.

Create and share

Make a colourful, labelled poster that teaches calculating with simple data. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.1.5

Lesson 16: Play & Challenge

PLAY — Choose one best answer for each question.

1. Calculating With Simple Data — Choose the correct name for something that tells how many items are present.

A. Count B. Average C. Sum D. Minimum

2. Calculating With Simple Data — Your friend says, "This adds numbers together." What is the friend describing?

A. Sum B. Average C. Maximum D. Count

3. Calculating With Simple Data — Which idea from calculating with simple data matches the clue "shares a total evenly across the number of values"?

A. Maximum B. Average C. Sum D. Count

4. Calculating With Simple Data — The teacher asks for an example that is the largest value. Which answer fits?

A. Minimum B. Count C. Sum D. Maximum

5. Calculating With Simple Data — Can you identify the term described here: is the smallest value?

A. Maximum B. Average C. Minimum D. Sum

CHALLENGE — Think and apply.

1. Calculating With Simple Data — Imagine helping a younger pupil understand something that shares a total evenly across the number of values. Which term would you teach?

A. Average B. Maximum C. Count D. Sum

2. Calculating With Simple Data — Which concept matches the explanation "is the smallest value" when you review calculating with simple data?

A. Average B. Sum C. Minimum D. Maximum

3. Calculating With Simple Data — Which answer would fit on a poster beside the words "tells how many items are present"?

A. Sum B. Average C. Count D. Minimum

4. Calculating With Simple Data — A group makes a display on calculating with simple data. Their first clue is "adds numbers together". Which label is correct?

A. Average B. Count C. Sum D. Maximum

5. Calculating With Simple Data — At home, Yaw remembers a class rule or concept that is the largest value. What was it called?

A. Sum B. Count C. Minimum D. Maximum

6. Calculating With Simple Data — For a short demonstration, the teacher needs an example that shares a total evenly across the number of values. Which choice would you select?

A. Average B. Sum C. Maximum D. Count

7. Calculating With Simple Data — Your class is solving a practical problem. One useful thing is the largest value. What is its name?

A. Sum B. Count C. Minimum D. Maximum

8. Calculating With Simple Data — During a school ICT lesson, Yaw wants to show something that tells how many items are present. Which idea should be labelled?

A. Minimum B. Sum C. Count D. Average

9. Calculating With Simple Data — At the community learning centre, someone describes a tool or idea that is the smallest value. Which one is it?

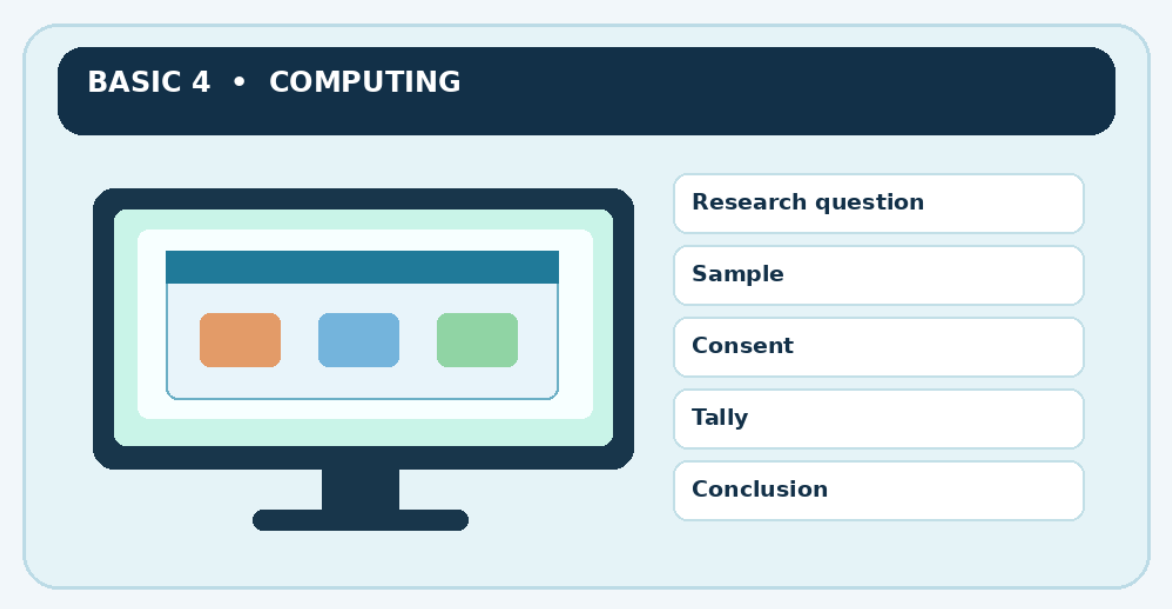
A. Average B. Sum C. Minimum D. Maximum

10. Calculating With Simple Data — A classmate confuses several ideas about calculating with simple data. Which one actually adds numbers together?

A. Sum B. Maximum C. Count D. Average

Lesson 17: Steps in Data Collection

We will learn to: Identify, explain and apply the important ideas behind steps in data collection.



Five important ideas

Research question: states what we want to discover.

Sample: is a smaller selected group.

Consent: means someone agrees to participate.

Tally: records counts using marks.

Conclusion: explains what the collected results suggest.

Our Ghanaian story

When interviewing classmates about favourite fruits, Kofi had a question about steps in data collection. The group discovered that research question states what we want to discover, while sample is a smaller selected group. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Research question, Sample and Consent; explain what each means.

Create and share

Make a colourful, labelled poster that teaches steps in data collection. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.2.1

Lesson 17: Play & Challenge

PLAY — Choose one best answer for each question.

1. Steps in Data Collection — Choose the correct name for something that states what we want to discover.

A. Conclusion B. Sample C. Research question D. Consent

2. Steps in Data Collection — Your friend says, "This is a smaller selected group." What is the friend describing?

A. Research question B. Consent C. Sample D. Conclusion

3. Steps in Data Collection — Which idea from steps in data collection matches the clue "means someone agrees to participate"?

A. Research question B. Tally C. Conclusion D. Consent

4. Steps in Data Collection — The teacher asks for an example that records counts using marks. Which answer fits?

A. Conclusion B. Sample C. Research question D. Tally

5. Steps in Data Collection — Can you identify the term described here: explains what the collected results suggest?

A. Tally B. Consent C. Sample D. Conclusion

CHALLENGE — Think and apply.

1. Steps in Data Collection — Imagine helping a younger pupil understand something that means someone agrees to participate. Which term would you teach?

A. Conclusion B. Research question C. Consent D. Tally

2. Steps in Data Collection — A group makes a display on steps in data collection. Their first clue is "is a smaller selected group". Which label is correct?

A. Consent B. Conclusion C. Sample D. Research question

3. Steps in Data Collection — Which answer would fit on a poster beside the words "states what we want to discover"?

A. Research question B. Conclusion C. Sample D. Consent

4. Steps in Data Collection — For a short demonstration, the teacher needs an example that means someone agrees to participate. Which choice would you select?

A. Tally B. Conclusion C. Research question D. Consent

5. Steps in Data Collection — Your class is solving a practical problem. One useful thing records counts using marks. What is its name?

A. Conclusion B. Tally C. Sample D. Research question

6. Steps in Data Collection — During a school ICT lesson, Kofi wants to show something that states what we want to discover. Which idea should be labelled?

A. Research question B. Consent C. Sample D. Conclusion

7. Steps in Data Collection — A classmate confuses several ideas about steps in data collection. Which one actually is a smaller selected group?

A. Sample B. Conclusion C. Consent D. Research question

8. Steps in Data Collection — Which concept matches the explanation "explains what the collected results suggest" when you review steps in data collection?

A. Conclusion B. Sample C. Tally D. Consent

9. Steps in Data Collection — At home, Kofi remembers a class rule or concept that records counts using marks. What was it called?

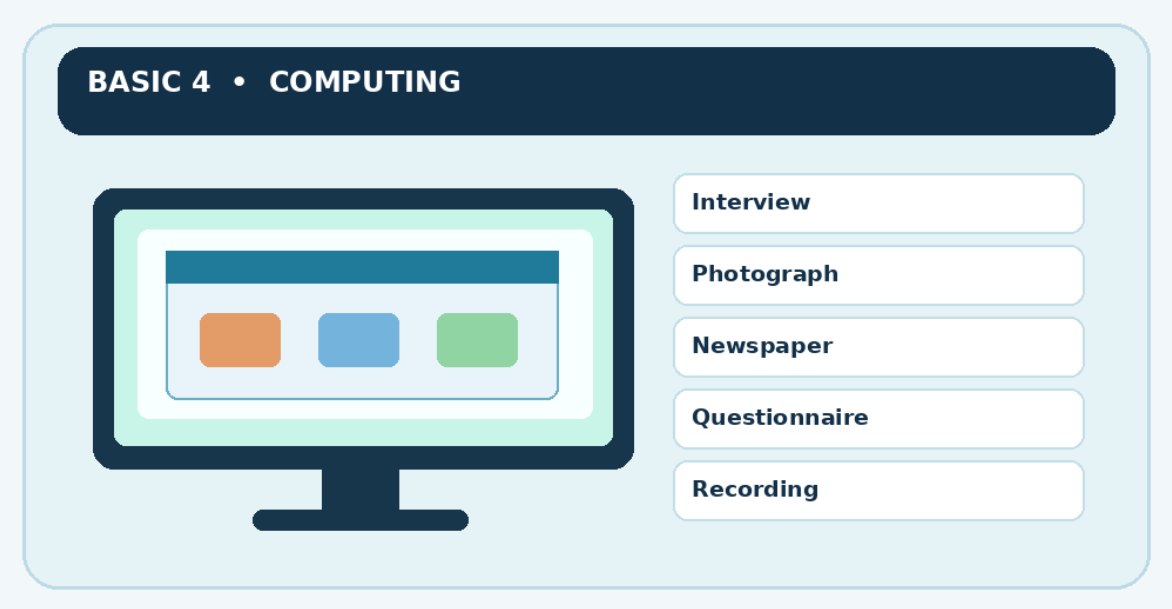
A. Research question B. Conclusion C. Tally D. Sample

10. Steps in Data Collection — At the community learning centre, someone describes a tool or idea that explains what the collected results suggest. Which one is it?

A. Consent B. Conclusion C. Sample D. Tally

Lesson 18: Data Collection Tools

We will learn to: Identify, explain and apply the important ideas behind data collection tools.



Five important ideas

Interview: collects answers by talking to people.

Photograph: records a visual moment.

Newspaper: reports news and events.

Questionnaire: collects written responses.

Recording: stores captured sound or video.

Our Ghanaian story

When preparing a report about the neighbourhood, Ama had a question about data collection tools. The group discovered that interview collects answers by talking to people, while photograph records a visual moment. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Interview, Photograph and Newspaper; explain what each means.

Create and share

Make a colourful, labelled poster that teaches data collection tools. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.2.2

Lesson 18: Play & Challenge

PLAY — Choose one best answer for each question.

1. Data Collection Tools — Choose the correct name for something that collects answers by talking to people.

A. Questionnaire B. Photograph C. Recording D. Interview

2. Data Collection Tools — Your friend says, "This records a visual moment." What is the friend describing?

A. Newspaper B. Photograph C. Questionnaire D. Recording

3. Data Collection Tools — Which idea from data collection tools matches the clue "reports news and events"?

A. Questionnaire B. Photograph C. Recording D. Newspaper

4. Data Collection Tools — The teacher asks for an example that collects written responses. Which answer fits?

A. Photograph B. Newspaper C. Questionnaire D. Interview

5. Data Collection Tools — Can you identify the term described here: stores captured sound or video?

A. Interview B. Questionnaire C. Newspaper D. Recording

CHALLENGE — Think and apply.

1. Data Collection Tools — At home, Akosua remembers a class rule or concept that collects written responses. What was it called?

A. Photograph B. Questionnaire C. Interview D. Newspaper

2. Data Collection Tools — Imagine helping a younger pupil understand something that reports news and events. Which term would you teach?

A. Photograph B. Newspaper C. Recording D. Questionnaire

3. Data Collection Tools — For a short demonstration, the teacher needs an example that reports news and events. Which choice would you select?

A. Newspaper B. Questionnaire C. Recording D. Photograph

4. Data Collection Tools — Which concept matches the explanation "stores captured sound or video" when you review data collection tools?

A. Recording B. Newspaper C. Questionnaire D. Interview

5. Data Collection Tools — During a school ICT lesson, Akosua wants to show something that collects answers by talking to people. Which idea should be labelled?

A. Photograph B. Recording C. Interview D. Questionnaire

6. Data Collection Tools — At the community learning centre, someone describes a tool or idea that stores captured sound or video. Which one is it?

A. Interview B. Recording C. Questionnaire D. Newspaper

7. Data Collection Tools — A classmate confuses several ideas about data collection tools. Which one actually records a visual moment?

A. Recording B. Photograph C. Questionnaire D. Newspaper

8. Data Collection Tools — Your class is solving a practical problem. One useful thing collects written responses. What is its name?

A. Newspaper B. Interview C. Questionnaire D. Photograph

9. Data Collection Tools — Which answer would fit on a poster beside the words "collects answers by talking to people"?

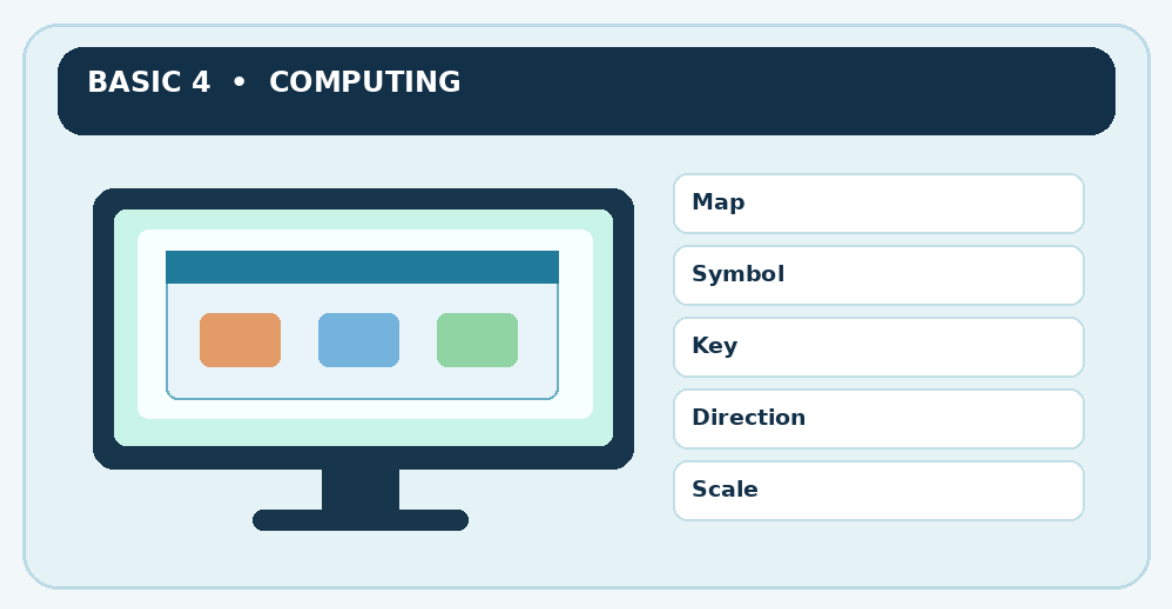
A. Photograph B. Interview C. Questionnaire D. Recording

10. Data Collection Tools — A group makes a display on data collection tools. Their first clue is "records a visual moment". Which label is correct?

A. Photograph B. Recording C. Questionnaire D. Newspaper

Lesson 19: Sketch a Simple Community Map

We will learn to: Identify, explain and apply the important ideas behind sketch a simple community map.



Five important ideas

Map: shows positions and places in an area.

Symbol: stands for a place or feature.

Key: explains what map symbols mean.

Direction: helps describe where to move.

Scale: shows the relation between map distance and real distance.

Our Ghanaian story

When drawing a plan of the school compound, Kojo had a question about sketch a simple community map. The group discovered that map shows positions and places in an area, while symbol stands for a place or feature. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Map, Symbol and Key; explain what each means.

Create and share

Make a colourful, labelled poster that teaches sketch a simple community map. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.3.2.3

Lesson 19: Play & Challenge

PLAY — Choose one best answer for each question.

1. Sketch a Simple Community Map — Choose the correct name for something that shows positions and places in an area.

A. Scale B. Direction C. Map D. Symbol

2. Sketch a Simple Community Map — Your friend says, "This stands for a place or feature." What is the friend describing?

A. Scale B. Symbol C. Map D. Direction

3. Sketch a Simple Community Map — Which idea from sketch a simple community map matches the clue "explains what map symbols mean"?

A. Key B. Direction C. Scale D. Map

4. Sketch a Simple Community Map — The teacher asks for an example that helps describe where to move. Which answer fits?

A. Direction B. Key C. Scale D. Map

5. Sketch a Simple Community Map — Can you identify the term described here: shows the relation between map distance and real distance?

A. Map B. Scale C. Key D. Symbol

CHALLENGE — Think and apply.

1. Sketch a Simple Community Map — Your class is solving a practical problem. One useful thing helps describe where to move. What is its name?

A. Direction B. Scale C. Map D. Key

2. Sketch a Simple Community Map — A group makes a display on sketch a simple community map. Their first clue is "stands for a place or feature". Which label is correct?

A. Symbol B. Map C. Scale D. Direction

3. Sketch a Simple Community Map — A classmate confuses several ideas about sketch a simple community map. Which one actually stands for a place or feature?

A. Symbol B. Direction C. Scale D. Map

4. Sketch a Simple Community Map — Which concept matches the explanation "shows the relation between map distance and real distance" when you review sketch a simple community map?

A. Map B. Scale C. Key D. Symbol

5. Sketch a Simple Community Map — For a short demonstration, the teacher needs an example that explains what map symbols mean. Which choice would you select?

A. Key B. Scale C. Map D. Direction

6. Sketch a Simple Community Map — Imagine helping a younger pupil understand something that explains what map symbols mean. Which term would you teach?

A. Scale B. Map C. Key D. Direction

7. Sketch a Simple Community Map — At the community learning centre, someone describes a tool or idea that shows the relation between map distance and real distance. Which one is it?

A. Map B. Key C. Symbol D. Scale

8. Sketch a Simple Community Map — Which answer would fit on a poster beside the words "shows positions and places in an area"?

A. Scale B. Direction C. Map D. Symbol

9. Sketch a Simple Community Map — At home, Esi remembers a class rule or concept that helps describe where to move. What was it called?

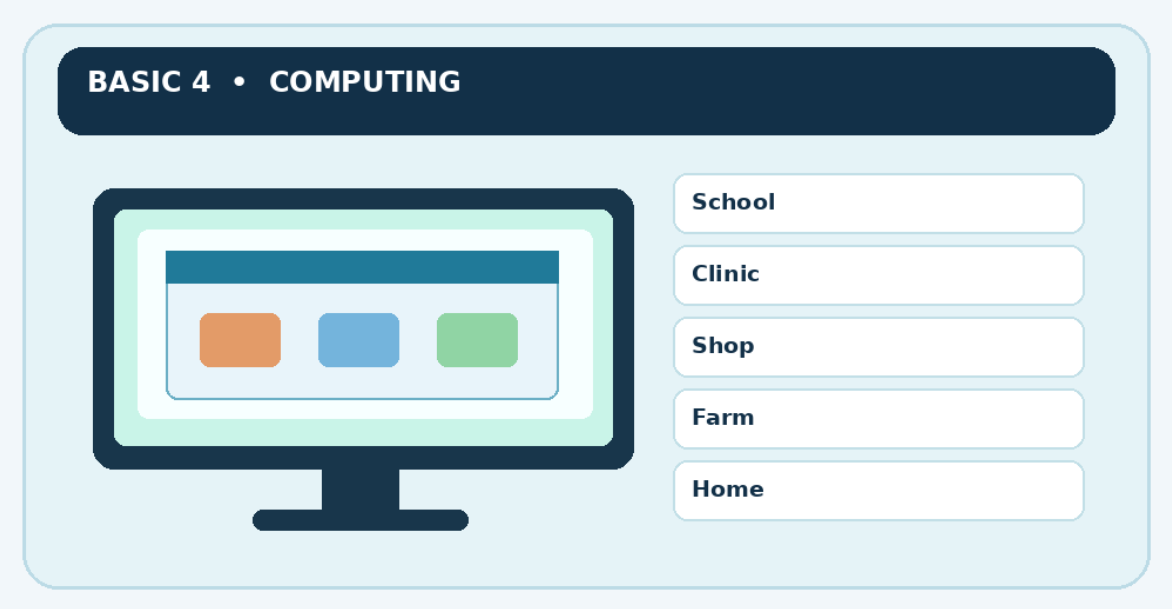
A. Direction B. Scale C. Key D. Map

10. Sketch a Simple Community Map — During a school ICT lesson, Esi wants to show something that shows positions and places in an area. Which idea should be labelled?

A. Scale B. Direction C. Map D. Symbol

Lesson 20: Digital Tools in the Community

We will learn to: Identify, explain and apply the important ideas behind digital tools in the community.



Five important ideas

School: uses computers for lessons and reading.

Clinic: uses computers for patient information.

Shop: uses computers for sales records.

Farm: uses technology for weather information.

Home: uses devices for learning and communication.

Our Ghanaian story

During a walk through the community, Esi had a question about digital tools in the community. The group discovered that school uses computers for lessons and reading, while clinic uses computers for patient information. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of school, clinic and shop. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches digital tools in the community. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.4.1.1

Lesson 20: Play & Challenge

PLAY — Choose one best answer for each question.

1. Digital Tools in the Community — Choose the correct name for something that uses computers for lessons and reading.

A. Farm B. Clinic C. School D. Home

2. Digital Tools in the Community — Your friend says, "This uses computers for patient information." What is the friend describing?

A. Shop B. School C. Home D. Clinic

3. Digital Tools in the Community — Which idea from digital tools in the community matches the clue "uses computers for sales records"?

A. Farm B. Home C. School D. Shop

4. Digital Tools in the Community — The teacher asks for an example that uses technology for weather information. Which answer fits?

A. Farm B. Shop C. Clinic D. Home

5. Digital Tools in the Community — Can you identify the term described here: uses devices for learning and communication?

A. Home B. Farm C. School D. Shop

CHALLENGE — Think and apply.

1. Digital Tools in the Community — At home, Ama remembers a class rule or concept that uses technology for weather information. What was it called?

A. Shop B. Home C. Farm D. Clinic

2. Digital Tools in the Community — Imagine helping a younger pupil understand something that uses computers for sales records. Which term would you teach?

A. Farm B. Home C. Shop D. School

3. Digital Tools in the Community — Your class is solving a practical problem. One useful thing uses technology for weather information. What is its name?

A. Farm B. Home C. Clinic D. Shop

4. Digital Tools in the Community — During a school ICT lesson, Ama wants to show something that uses computers for lessons and reading. Which idea should be labelled?

A. Clinic B. Home C. Farm D. School

5. Digital Tools in the Community — Which concept matches the explanation "uses devices for learning and communication" when you review digital tools in the community?

A. Farm B. Shop C. Home D. School

6. Digital Tools in the Community — Which answer would fit on a poster beside the words "uses computers for lessons and reading"?

A. School B. Home C. Farm D. Clinic

7. Digital Tools in the Community — At the community learning centre, someone describes a tool or idea that uses devices for learning and communication. Which one is it?

A. Shop B. Home C. Farm D. School

8. Digital Tools in the Community — A group makes a display on digital tools in the community. Their first clue is "uses computers for patient information". Which label is correct?

A. Clinic B. School C. Home D. Shop

9. Digital Tools in the Community — For a short demonstration, the teacher needs an example that uses computers for sales records. Which choice would you select?

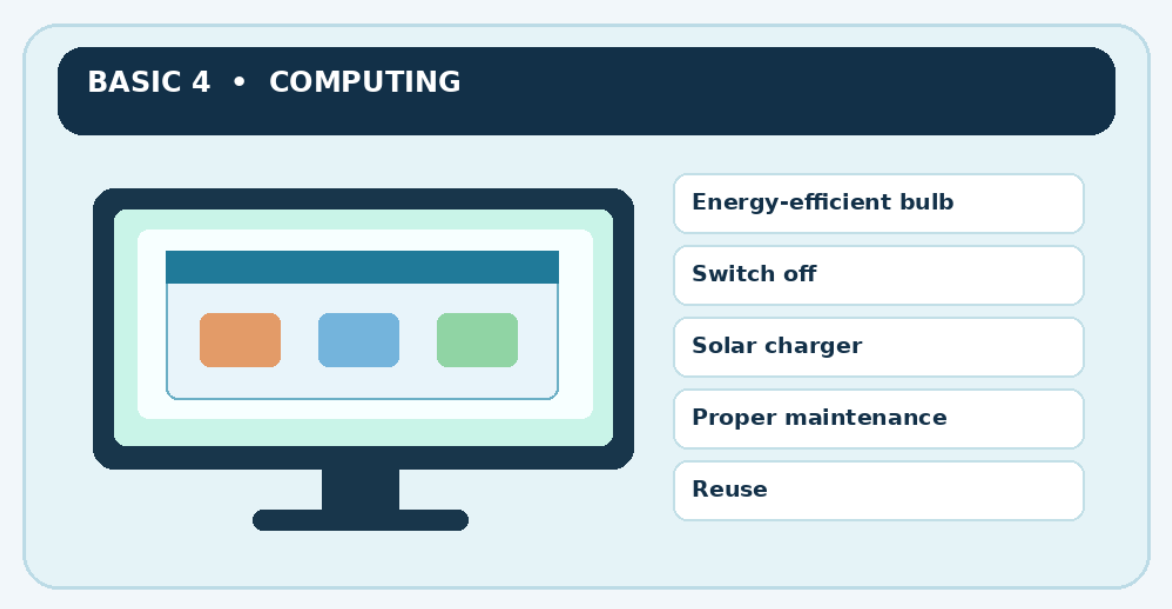
A. Farm B. Shop C. School D. Home

10. Digital Tools in the Community — A classmate confuses several ideas about digital tools in the community. Which one actually uses computers for patient information?

A. School B. Clinic C. Home D. Shop

Lesson 21: Energy-Saving Technology

We will learn to: Identify, explain and apply the important ideas behind energy-saving technology.



Five important ideas

Energy-efficient bulb: uses less electricity for similar light.

Switch off: saves power when a device is not needed.

Solar charger: uses energy from sunlight.

Proper maintenance: helps devices work efficiently.

Reuse: can reduce waste from usable equipment.

Our Ghanaian story

During an energy-saving campaign at school, Yaw had a question about energy-saving technology. The group discovered that energy-efficient bulb uses less electricity for similar light, while switch off saves power when a device is not needed. They drew an example and explained it to their classmates.

Try it!

Create a class safety demonstration. Show a good example of energy-efficient bulb and switch off and explain why the good practice matters.

Create and share

Make a colourful, labelled poster that teaches energy-saving technology. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.4.1.2

Lesson 21: Play & Challenge

PLAY — Choose one best answer for each question.

1. Energy-Saving Technology — Choose the correct name for something that uses less electricity for similar light.

A. Solar charger B. Reuse C. Energy-efficient bulb D. Switch off

2. Energy-Saving Technology — Your friend says, "This saves power when a device is not needed." What is the friend describing?

A. Proper maintenance B. Reuse C. Energy-efficient bulb D. Switch off

3. Energy-Saving Technology — Which idea from energy-saving technology matches the clue "uses energy from sunlight"?

A. Energy-efficient bulb B. Proper maintenance C. Reuse D. Solar charger

4. Energy-Saving Technology — The teacher asks for an example that helps devices work efficiently. Which answer fits?

A. Solar charger B. Proper maintenance C. Reuse D. Energy-efficient bulb

5. Energy-Saving Technology — Can you identify the term described here: can reduce waste from usable equipment?

A. Reuse B. Proper maintenance C. Switch off D. Solar charger

CHALLENGE — Think and apply.

1. Energy-Saving Technology — A group makes a display on energy-saving technology. Their first clue is "saves power when a device is not needed". Which label is correct?

A. Proper maintenance B. Energy-efficient bulb C. Reuse D. Switch off

2. Energy-Saving Technology — Imagine helping a younger pupil understand something that uses energy from sunlight. Which term would you teach?

A. Proper maintenance B. Energy-efficient bulb C. Reuse D. Solar charger

3. Energy-Saving Technology — Which concept matches the explanation "can reduce waste from usable equipment" when you review energy-saving technology?

A. Reuse B. Switch off C. Proper maintenance D. Solar charger

4. Energy-Saving Technology — A classmate confuses several ideas about energy-saving technology. Which one actually saves power when a device is not needed?

A. Energy-efficient bulb B. Switch off C. Reuse D. Proper maintenance

5. Energy-Saving Technology — Your class is solving a practical problem. One useful thing helps devices work efficiently. What is its name?

A. Solar charger B. Proper maintenance C. Energy-efficient bulb D. Reuse

6. Energy-Saving Technology — During a school ICT lesson, Yaw wants to show something that uses less electricity for similar light. Which idea should be labelled?

A. Switch off B. Energy-efficient bulb C. Solar charger D. Reuse

7. Energy-Saving Technology — At the community learning centre, someone describes a tool or idea that can reduce waste from usable equipment. Which one is it?

A. Switch off B. Solar charger C. Reuse D. Proper maintenance

8. Energy-Saving Technology — At home, Yaw remembers a class rule or concept that helps devices work efficiently. What was it called?

A. Solar charger B. Proper maintenance C. Reuse D. Energy-efficient bulb

9. Energy-Saving Technology — For a short demonstration, the teacher needs an example that uses energy from sunlight. Which choice would you select?

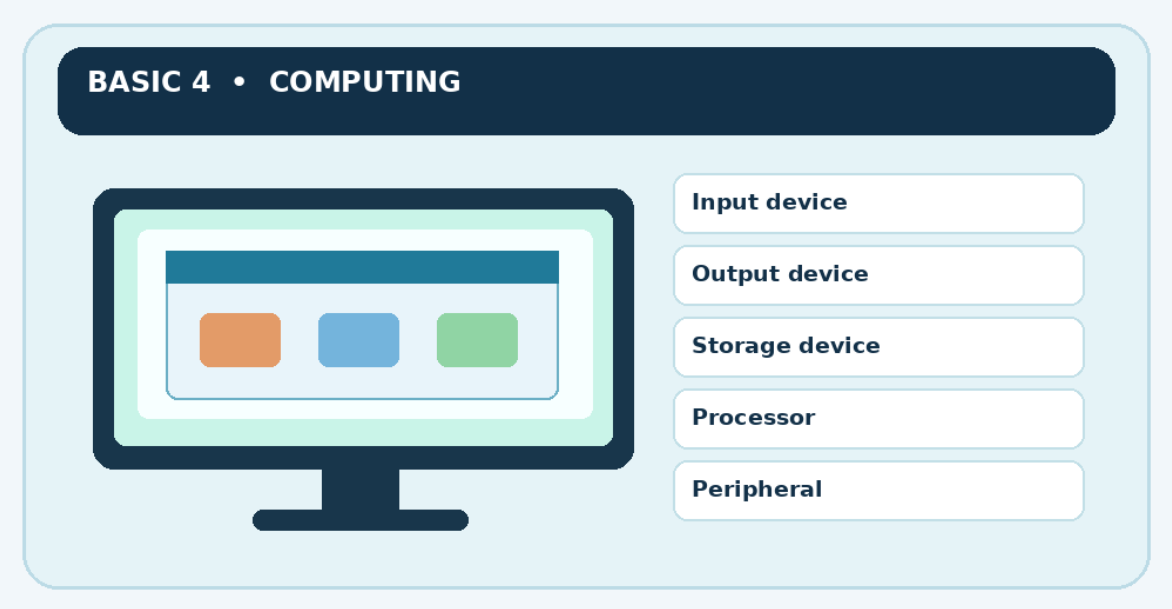
A. Solar charger B. Reuse C. Energy-efficient bulb D. Proper maintenance

10. Energy-Saving Technology — Which answer would fit on a poster beside the words "uses less electricity for similar light"?

A. Energy-efficient bulb B. Reuse C. Solar charger D. Switch off

Lesson 22: Digital Devices and Peripherals

We will learn to: Identify, explain and apply the important ideas behind digital devices and peripherals.



Five important ideas

Input device: sends information into a computer.

Output device: shows or plays results.

Storage device: keeps digital files.

Processor: helps follow computing instructions.

Peripheral: is an extra device connected to a computer.

Our Ghanaian story

During a tour of the computer room, Abena had a question about digital devices and peripherals. The group discovered that input device sends information into a computer, while output device shows or plays results. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of input device, output device and storage device. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches digital devices and peripherals. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.4.1.3

Lesson 22: Play & Challenge

PLAY — Choose one best answer for each question.

1. Digital Devices and Peripherals — Choose the correct name for something that sends information into a computer.

A. Output device B. Processor C. Storage device D. Input device

2. Digital Devices and Peripherals — Your friend says, "This shows or plays results." What is the friend describing?

A. Output device B. Peripheral C. Storage device D. Input device

3. Digital Devices and Peripherals — Which idea from digital devices and peripherals matches the clue "keeps digital files"?

A. Input device B. Storage device C. Processor D. Peripheral

4. Digital Devices and Peripherals — The teacher asks for an example that helps follow computing instructions. Which answer fits?

A. Input device B. Peripheral C. Processor D. Output device

5. Digital Devices and Peripherals — Can you identify the term described here: is an extra device connected to a computer?

A. Peripheral B. Processor C. Storage device D. Input device

CHALLENGE — Think and apply.

1. Digital Devices and Peripherals — Which answer would fit on a poster beside the words "sends information into a computer"?

A. Storage device B. Processor C. Output device D. Input device

2. Digital Devices and Peripherals — A group makes a display on digital devices and peripherals. Their first clue is "shows or plays results". Which label is correct?

A. Peripheral B. Storage device C. Input device D. Output device

3. Digital Devices and Peripherals — At home, Kofi remembers a class rule or concept that helps follow computing instructions. What was it called?

A. Input device B. Output device C. Processor D. Peripheral

4. Digital Devices and Peripherals — For a short demonstration, the teacher needs an example that keeps digital files. Which choice would you select?

A. Processor B. Storage device C. Input device D. Peripheral

5. Digital Devices and Peripherals — A classmate confuses several ideas about digital devices and peripherals. Which one actually shows or plays results?

A. Peripheral B. Storage device C. Input device D. Output device

6. Digital Devices and Peripherals — Your class is solving a practical problem. One useful thing helps follow computing instructions. What is its name?

A. Output device B. Input device C. Peripheral D. Processor

7. Digital Devices and Peripherals — Which concept matches the explanation "is an extra device connected to a computer" when you review digital devices and peripherals?

A. Input device B. Storage device C. Peripheral D. Processor

8. Digital Devices and Peripherals — Imagine helping a younger pupil understand something that keeps digital files. Which term would you teach?

A. Processor B. Input device C. Storage device D. Peripheral

9. Digital Devices and Peripherals — During a school ICT lesson, Kofi wants to show something that sends information into a computer. Which idea should be labelled?

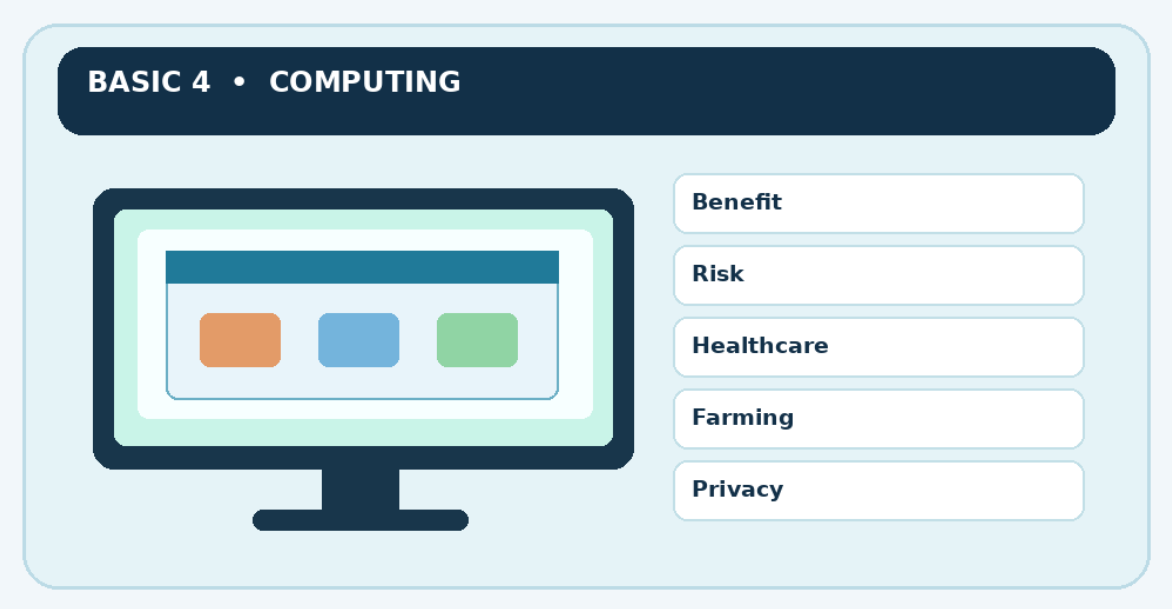
A. Processor B. Storage device C. Input device D. Output device

10. Digital Devices and Peripherals — At the community learning centre, someone describes a tool or idea that is an extra device connected to a computer. Which one is it?

A. Peripheral B. Input device C. Processor D. Storage device

Lesson 23: Changing Technology Around Us

We will learn to: Identify, explain and apply the important ideas behind changing technology around us.



Five important ideas

Benefit: is a helpful effect of technology.

Risk: is a possible harmful result.

Healthcare: can use technology to support patients.

Farming: can use sensors and weather data.

Privacy: means controlling access to personal information.

Our Ghanaian story

While discussing changes in a Ghanaian community, Kofi had a question about changing technology around us. The group discovered that benefit is a helpful effect of technology, while risk is a possible harmful result. They drew an example and explained it to their classmates.

Try it!

Create a class safety demonstration. Show a good example of benefit and risk and explain why the good practice matters.

Create and share

Make a colourful, labelled poster that teaches changing technology around us. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.1.4.1.4

Lesson 23: Play & Challenge

PLAY — Choose one best answer for each question.

1. Changing Technology Around Us — Choose the correct name for something that is a helpful effect of technology.

A. Benefit B. Healthcare C. Privacy D. Farming

2. Changing Technology Around Us — Your friend says, "This is a possible harmful result." What is the friend describing?

A. Farming B. Benefit C. Risk D. Privacy

3. Changing Technology Around Us — Which idea from changing technology around us matches the clue "can use technology to support patients"?

A. Benefit B. Risk C. Healthcare D. Farming

4. Changing Technology Around Us — The teacher asks for an example that can use sensors and weather data. Which answer fits?

A. Benefit B. Risk C. Farming D. Healthcare

5. Changing Technology Around Us — Can you identify the term described here: means controlling access to personal information?

A. Risk B. Benefit C. Farming D. Privacy

CHALLENGE — Think and apply.

1. Changing Technology Around Us — Imagine helping a younger pupil understand something that can use technology to support patients. Which term would you teach?

A. Healthcare B. Farming C. Risk D. Benefit

2. Changing Technology Around Us — A classmate confuses several ideas about changing technology around us. Which one actually is a possible harmful result?

A. Farming B. Privacy C. Risk D. Benefit

3. Changing Technology Around Us — A group makes a display on changing technology around us. Their first clue is "is a possible harmful result". Which label is correct?

A. Farming B. Privacy C. Risk D. Benefit

4. Changing Technology Around Us — At home, Akosua remembers a class rule or concept that can use sensors and weather data. What was it called?

A. Benefit B. Risk C. Healthcare D. Farming

5. Changing Technology Around Us — Which concept matches the explanation "means controlling access to personal information" when you review changing technology around us?

A. Privacy B. Benefit C. Farming D. Risk

6. Changing Technology Around Us — During a school ICT lesson, Akosua wants to show something that is a helpful effect of technology. Which idea should be labelled?

A. Healthcare B. Farming C. Privacy D. Benefit

7. Changing Technology Around Us — Which answer would fit on a poster beside the words "is a helpful effect of technology"?

A. Healthcare B. Farming C. Benefit D. Privacy

8. Changing Technology Around Us — For a short demonstration, the teacher needs an example that can use technology to support patients. Which choice would you select?

A. Benefit B. Farming C. Risk D. Healthcare

9. Changing Technology Around Us — At the community learning centre, someone describes a tool or idea that means controlling access to personal information. Which one is it?

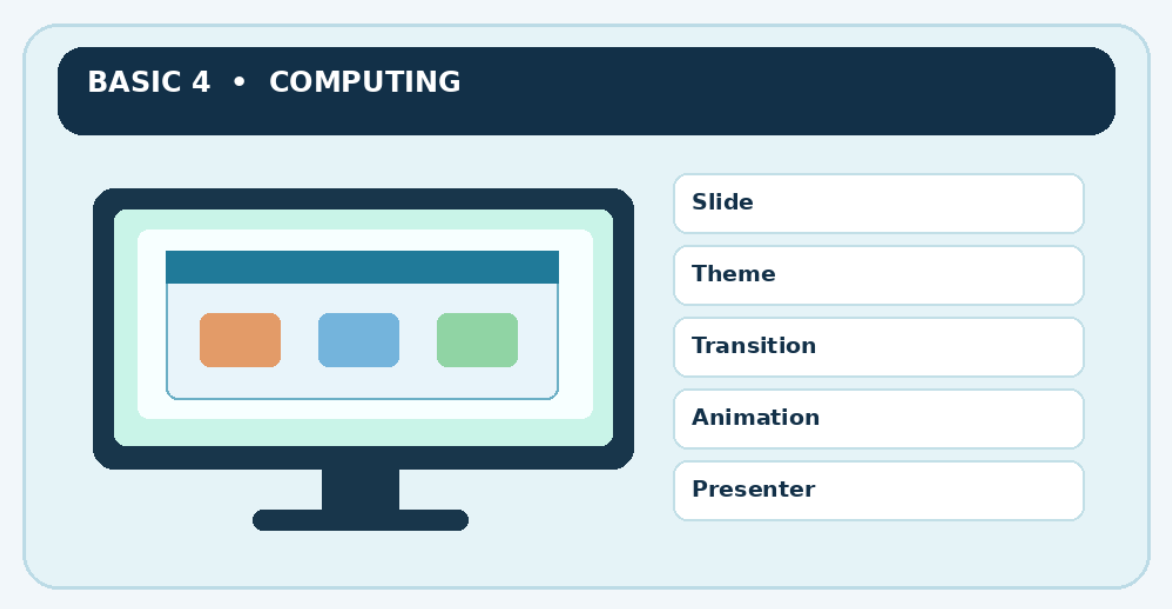
A. Risk B. Benefit C. Privacy D. Farming

10. Changing Technology Around Us — Your class is solving a practical problem. One useful thing can use sensors and weather data. What is its name?

A. Farming B. Risk C. Benefit D. Healthcare

Lesson 24: PowerPoint Home Tools

We will learn to: Identify, explain and apply the important ideas behind powerpoint home tools.



Five important ideas

Slide: is one screen of a presentation.

Theme: gives slides a coordinated design.

Transition: controls how one slide changes to another.

Animation: makes an object appear or move.

Presenter: speaks clearly while showing the slides.

Our Ghanaian story

While preparing a school assembly talk, Ama had a question about powerpoint home tools. The group discovered that slide is one screen of a presentation, while theme gives slides a coordinated design. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include slide, theme and transition; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches powerpoint home tools. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.2.1.1.1

Lesson 24: Play & Challenge

PLAY — Choose one best answer for each question.

1. PowerPoint Home Tools — Choose the correct name for something that is one screen of a presentation.

A. Slide B. Presenter C. Animation D. Transition

2. PowerPoint Home Tools — Your friend says, "This gives slides a coordinated design." What is the friend describing?

A. Animation B. Slide C. Theme D. Presenter

3. PowerPoint Home Tools — Which idea from powerpoint home tools matches the clue "controls how one slide changes to another"?

A. Theme B. Presenter C. Transition D. Animation

4. PowerPoint Home Tools — The teacher asks for an example that makes an object appear or move. Which answer fits?

A. Slide B. Transition C. Theme D. Animation

5. PowerPoint Home Tools — Can you identify the term described here: speaks clearly while showing the slides?

A. Transition B. Theme C. Slide D. Presenter

CHALLENGE — Think and apply.

1. PowerPoint Home Tools — A classmate confuses several ideas about powerpoint home tools. Which one actually gives slides a coordinated design?

A. Presenter B. Animation C. Theme D. Slide

2. PowerPoint Home Tools — At the community learning centre, someone describes a tool or idea that speaks clearly while showing the slides. Which one is it?

A. Transition B. Slide C. Presenter D. Theme

3. PowerPoint Home Tools — Your class is solving a practical problem. One useful thing makes an object appear or move. What is its name?

A. Animation B. Theme C. Transition D. Slide

4. PowerPoint Home Tools — Which concept matches the explanation "speaks clearly while showing the slides" when you review powerpoint home tools?

A. Presenter B. Theme C. Transition D. Slide

5. PowerPoint Home Tools — Imagine helping a younger pupil understand something that controls how one slide changes to another. Which term would you teach?

A. Transition B. Presenter C. Animation D. Theme

6. PowerPoint Home Tools — At home, Esi remembers a class rule or concept that makes an object appear or move. What was it called?

A. Slide B. Theme C. Transition D. Animation

7. PowerPoint Home Tools — For a short demonstration, the teacher needs an example that controls how one slide changes to another. Which choice would you select?

A. Transition B. Presenter C. Animation D. Theme

8. PowerPoint Home Tools — Which answer would fit on a poster beside the words "is one screen of a presentation"?

A. Animation B. Transition C. Slide D. Presenter

9. PowerPoint Home Tools — During a school ICT lesson, Esi wants to show something that is one screen of a presentation. Which idea should be labelled?

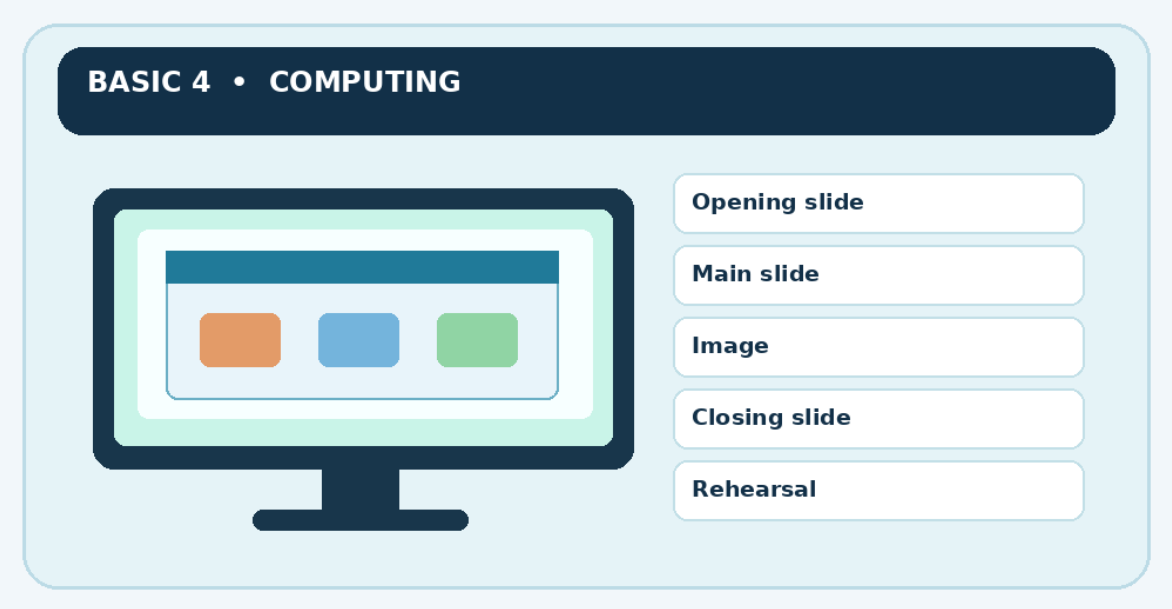
A. Slide B. Transition C. Presenter D. Animation

10. PowerPoint Home Tools — A group makes a display on powerpoint home tools. Their first clue is "gives slides a coordinated design". Which label is correct?

A. Animation B. Presenter C. Theme D. Slide

Lesson 25: Make a Three-Slide Presentation

We will learn to: Identify, explain and apply the important ideas behind make a three-slide presentation.



Five important ideas

Opening slide: introduces the topic.

Main slide: shows one key idea.

Image: helps communicate meaning.

Closing slide: sums up the message.

Rehearsal: helps a presenter speak with confidence.

Our Ghanaian story

Before presenting an environmental club project, Kojo had a question about make a three-slide presentation. The group discovered that opening slide introduces the topic, while main slide shows one key idea. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include opening slide, main slide and image; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches make a three-slide presentation. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.2.1.1.2

Lesson 25: Play & Challenge

PLAY — Choose one best answer for each question.

1. Make a Three-Slide Presentation — Choose the correct name for something that introduces the topic.

A. Opening slide B. Image C. Closing slide D. Rehearsal

2. Make a Three-Slide Presentation — Your friend says, "This shows one key idea." What is the friend describing?

A. Opening slide B. Main slide C. Closing slide D. Image

3. Make a Three-Slide Presentation — Which idea from make a three-slide presentation matches the clue "helps communicate meaning"?

A. Image B. Opening slide C. Rehearsal D. Main slide

4. Make a Three-Slide Presentation — The teacher asks for an example that sums up the message. Which answer fits?

A. Closing slide B. Image C. Opening slide D. Rehearsal

5. Make a Three-Slide Presentation — Can you identify the term described here: helps a presenter speak with confidence?

A. Opening slide B. Rehearsal C. Main slide D. Image

CHALLENGE — Think and apply.

1. Make a Three-Slide Presentation — For a short demonstration, the teacher needs an example that helps communicate meaning. Which choice would you select?

A. Main slide B. Opening slide C. Image D. Rehearsal

2. Make a Three-Slide Presentation — Your class is solving a practical problem. One useful thing sums up the message. What is its name?

A. Image B. Opening slide C. Rehearsal D. Closing slide

3. Make a Three-Slide Presentation — Which concept matches the explanation "helps a presenter speak with confidence" when you review make a three-slide presentation?

A. Rehearsal B. Image C. Main slide D. Opening slide

4. Make a Three-Slide Presentation — During a school ICT lesson, Ama wants to show something that introduces the topic. Which idea should be labelled?

A. Image B. Rehearsal C. Opening slide D. Closing slide

5. Make a Three-Slide Presentation — A group makes a display on make a three-slide presentation. Their first clue is "shows one key idea". Which label is correct?

A. Closing slide B. Main slide C. Image D. Opening slide

6. Make a Three-Slide Presentation — Imagine helping a younger pupil understand something that helps communicate meaning. Which term would you teach?

A. Main slide B. Image C. Rehearsal D. Opening slide

7. Make a Three-Slide Presentation — At the community learning centre, someone describes a tool or idea that helps a presenter speak with confidence. Which one is it?

A. Image B. Opening slide C. Main slide D. Rehearsal

8. Make a Three-Slide Presentation — A classmate confuses several ideas about make a three-slide presentation. Which one actually shows one key idea?

A. Opening slide B. Main slide C. Closing slide D. Image

9. Make a Three-Slide Presentation — Which answer would fit on a poster beside the words "introduces the topic"?

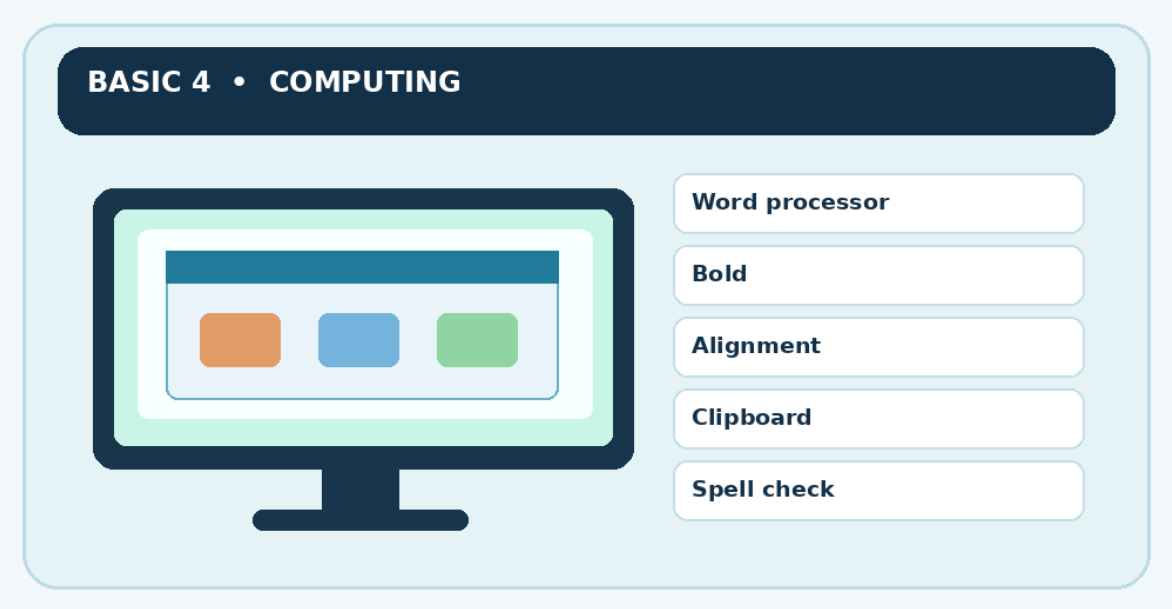
A. Opening slide B. Closing slide C. Rehearsal D. Image

10. Make a Three-Slide Presentation — At home, Ama remembers a class rule or concept that sums up the message. What was it called?

A. Opening slide B. Rehearsal C. Closing slide D. Image

Lesson 26: Word Processor Home Tools

We will learn to: Identify, explain and apply the important ideas behind word processor home tools.



Five important ideas

Word processor: creates and edits text documents.

Bold: makes important words stand out.

Alignment: positions text on the page.

Clipboard: temporarily holds copied material.

Spell check: helps find possible spelling mistakes.

Our Ghanaian story

While typing a school announcement, Esi had a question about word processor home tools. The group discovered that word processor creates and edits text documents, while bold makes important words stand out. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include word processor, bold and alignment; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches word processor home tools. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.3.1.1.1

Lesson 26: Play & Challenge

PLAY — Choose one best answer for each question.

1. Word Processor Home Tools — Choose the correct name for something that creates and edits text documents.

A. Spell check B. Bold C. Alignment D. Word processor

2. Word Processor Home Tools — Your friend says, "This makes important words stand out." What is the friend describing?

A. Bold B. Clipboard C. Alignment D. Word processor

3. Word Processor Home Tools — Which idea from word processor home tools matches the clue "positions text on the page"?

A. Alignment B. Word processor C. Spell check D. Bold

4. Word Processor Home Tools — The teacher asks for an example that temporarily holds copied material. Which answer fits?

A. Clipboard B. Alignment C. Bold D. Word processor

5. Word Processor Home Tools — Can you identify the term described here: helps find possible spelling mistakes?

A. Word processor B. Spell check C. Bold D. Alignment

CHALLENGE — Think and apply.

1. Word Processor Home Tools — A group makes a display on word processor home tools. Their first clue is "makes important words stand out". Which label is correct?

A. Clipboard B. Bold C. Word processor D. Alignment

2. Word Processor Home Tools — At the community learning centre, someone describes a tool or idea that helps find possible spelling mistakes. Which one is it?

A. Alignment B. Spell check C. Bold D. Word processor

3. Word Processor Home Tools — At home, Yaw remembers a class rule or concept that temporarily holds copied material. What was it called?

A. Word processor B. Alignment C. Bold D. Clipboard

4. Word Processor Home Tools — A classmate confuses several ideas about word processor home tools. Which one actually makes important words stand out?

A. Alignment B. Word processor C. Clipboard D. Bold

5. Word Processor Home Tools — Which answer would fit on a poster beside the words "creates and edits text documents"?

A. Spell check B. Alignment C. Word processor D. Bold

6. Word Processor Home Tools — For a short demonstration, the teacher needs an example that positions text on the page. Which choice would you select?

A. Bold B. Word processor C. Spell check D. Alignment

7. Word Processor Home Tools — During a school ICT lesson, Yaw wants to show something that creates and edits text documents. Which idea should be labelled?

A. Bold B. Spell check C. Word processor D. Alignment

8. Word Processor Home Tools — Which concept matches the explanation "helps find possible spelling mistakes" when you review word processor home tools?

A. Alignment B. Word processor C. Bold D. Spell check

9. Word Processor Home Tools — Imagine helping a younger pupil understand something that positions text on the page. Which term would you teach?

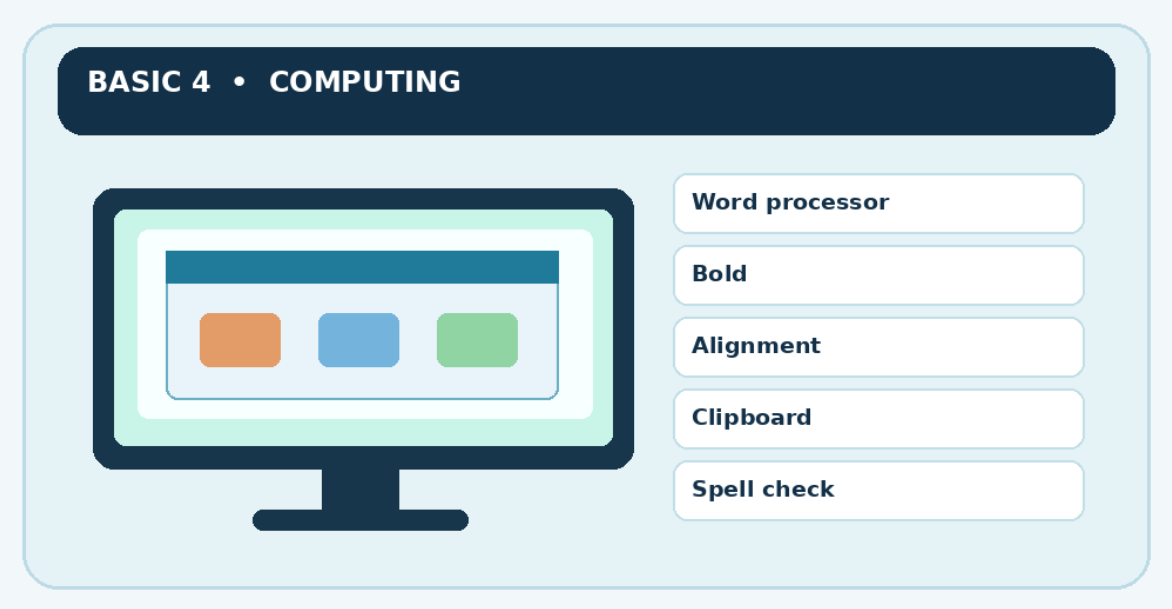
A. Alignment B. Bold C. Spell check D. Word processor

10. Word Processor Home Tools — Your class is solving a practical problem. One useful thing temporarily holds copied material. What is its name?

A. Word processor B. Bold C. Alignment D. Clipboard

Lesson 27: Format a Simple Document

We will learn to: Identify, explain and apply the important ideas behind format a simple document.



Five important ideas

Word processor: creates and edits text documents.

Bold: makes important words stand out.

Alignment: positions text on the page.

Clipboard: temporarily holds copied material.

Spell check: helps find possible spelling mistakes.

Our Ghanaian story

While typing a school announcement, Yaw had a question about format a simple document. The group discovered that word processor creates and edits text documents, while bold makes important words stand out. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include word processor, bold and alignment; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches format a simple document. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B4.3.1.1.2

Lesson 27: Play & Challenge

PLAY — Choose one best answer for each question.

1. Format a Simple Document — Choose the correct name for something that creates and edits text documents.

A. Word processor B. Alignment C. Spell check D. Clipboard

2. Format a Simple Document — Your friend says, "This makes important words stand out." What is the friend describing?

A. Alignment B. Spell check C. Bold D. Clipboard

3. Format a Simple Document — Which idea from format a simple document matches the clue "positions text on the page"?

A. Spell check B. Bold C. Alignment D. Clipboard

4. Format a Simple Document — The teacher asks for an example that temporarily holds copied material. Which answer fits?

A. Bold B. Alignment C. Spell check D. Clipboard

5. Format a Simple Document — Can you identify the term described here: helps find possible spelling mistakes?

A. Bold B. Alignment C. Word processor D. Spell check

CHALLENGE — Think and apply.

1. Format a Simple Document — A group makes a display on format a simple document. Their first clue is "makes important words stand out". Which label is correct?

A. Bold B. Clipboard C. Alignment D. Spell check

2. Format a Simple Document — Imagine helping a younger pupil understand something that positions text on the page. Which term would you teach?

A. Clipboard B. Bold C. Alignment D. Spell check

3. Format a Simple Document — During a school ICT lesson, Kofi wants to show something that creates and edits text documents. Which idea should be labelled?

A. Clipboard B. Word processor C. Alignment D. Spell check

4. Format a Simple Document — At home, Kofi remembers a class rule or concept that temporarily holds copied material. What was it called?

A. Bold B. Spell check C. Alignment D. Clipboard

5. Format a Simple Document — A classmate confuses several ideas about format a simple document. Which one actually makes important words stand out?

A. Spell check B. Bold C. Alignment D. Clipboard

6. Format a Simple Document — Which answer would fit on a poster beside the words "creates and edits text documents"?

A. Alignment B. Spell check C. Word processor D. Clipboard

7. Format a Simple Document — For a short demonstration, the teacher needs an example that positions text on the page. Which choice would you select?

A. Spell check B. Clipboard C. Alignment D. Bold

8. Format a Simple Document — Which concept matches the explanation "helps find possible spelling mistakes" when you review format a simple document?

A. Bold B. Spell check C. Alignment D. Word processor

9. Format a Simple Document — At the community learning centre, someone describes a tool or idea that helps find possible spelling mistakes. Which one is it?

A. Alignment B. Spell check C. Word processor D. Bold

10. Format a Simple Document — Your class is solving a practical problem. One useful thing temporarily holds copied material. What is its name?

A. Bold B. Clipboard C. Spell check D. Alignment

Teacher answer guide

Review the correct term and ask each learner to explain why it fits.

Lesson 1 — Computer Components

Play: 1. Monitor | 2. Keyboard | 3. Mouse | 4. System unit | 5. Speakers

Challenge: 1. System unit | 2. Keyboard | 3. Mouse | 4. Speakers | 5. Speakers | 6. System unit | 7. Monitor | 8. Monitor | 9. Mouse | 10. Keyboard

Lesson 2 — Different Input Devices

Play: 1. Keyboard | 2. Mouse | 3. Microphone | 4. Scanner | 5. Camera

Challenge: 1. Camera | 2. Mouse | 3. Keyboard | 4. Camera | 5. Microphone | 6. Mouse | 7. Microphone | 8. Keyboard | 9. Scanner | 10. Scanner

Lesson 3 — RAM and ROM

Play: 1. RAM | 2. ROM | 3. Storage | 4. Flash drive | 5. Hard drive

Challenge: 1. ROM | 2. Flash drive | 3. ROM | 4. RAM | 5. Hard drive | 6. RAM | 7. Storage | 8. Hard drive | 9. Storage | 10. Flash drive

Lesson 4 — Typing Short Phrases

Play: 1. Home row | 2. Top row | 3. Bottom row | 4. Practice | 5. Posture

Challenge: 1. Home row | 2. Posture | 3. Top row | 4. Bottom row | 5. Posture | 6. Home row | 7. Top row | 8. Practice | 9. Practice | 10. Bottom row

Lesson 5 — First Generation Computers

Play: 1. Vacuum tube | 2. Transistor | 3. Integrated circuit | 4. Microprocessor | 5. Personal computer

Challenge: 1. Integrated circuit | 2. Personal computer | 3. Transistor | 4. Transistor | 5. Microprocessor | 6. Microprocessor | 7. Integrated circuit | 8. Vacuum tube | 9. Vacuum tube | 10. Personal computer

Lesson 6 — Desktop Features

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Desktop | 2. Wallpaper | 3. Desktop | 4. Taskbar | 5. Wallpaper | 6. Start menu | 7. Start menu | 8. Taskbar | 9. Icon | 10. Icon

Lesson 7 — Personalise the Desktop

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Wallpaper | 2. Icon | 3. Icon | 4. Desktop | 5. Wallpaper | 6. Taskbar | 7. Taskbar | 8. Start menu | 9. Desktop | 10. Start menu

Lesson 8 — Taskbar and Start Menu

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Desktop | 2. Taskbar | 3. Icon | 4. Wallpaper | 5. Start menu | 6. Start menu | 7. Icon | 8. Desktop | 9. Wallpaper | 10. Taskbar

Lesson 9 — Add and Remove Desktop Icons

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Start menu | 2. Taskbar | 3. Desktop | 4. Desktop | 5. Icon | 6. Wallpaper | 7. Start menu | 8. Taskbar | 9. Wallpaper | 10. Icon

Lesson 10 — Moving and Selecting Icons

Play: 1. Single click | 2. Double click | 3. Drag | 4. Scroll wheel | 5. Right click

Challenge: 1. Drag | 2. Single click | 3. Double click | 4. Single click | 5. Right click | 6. Drag | 7. Scroll wheel | 8. Scroll wheel | 9. Double click | 10. Right click

Lesson 11 — Creating and Renaming Folders

Play: 1. Folder | 2. File name | 3. Copy | 4. Move | 5. Rename

Challenge: 1. Rename | 2. Folder | 3. Move | 4. File name | 5. Rename | 6. Copy | 7. File name | 8. Copy | 9. Folder | 10. Move

Lesson 12 — Identify Data in Everyday Reports

Play: 1. Data | 2. Information | 3. Survey | 4. Observation | 5. Table

Challenge: 1. Observation | 2. Table | 3. Observation | 4. Table | 5. Survey | 6. Survey | 7. Data | 8. Information | 9. Information | 10. Data

Lesson 13 — Classify Data into Information

Play: 1. Sort | 2. Category | 3. Label | 4. Pattern | 5. Chart

Challenge: 1. Pattern | 2. Pattern | 3. Label | 4. Label | 5. Sort | 6. Category | 7. Sort | 8. Chart | 9. Chart | 10. Category

Lesson 14 — Recognising Information Sources

Play: 1. Interview | 2. Photograph | 3. Newspaper | 4. Questionnaire | 5. Recording

Challenge: 1. Questionnaire | 2. Photograph | 3. Recording | 4. Photograph | 5. Interview | 6. Questionnaire | 7. Newspaper | 8. Newspaper | 9. Interview | 10. Recording

Lesson 15 — Sharing Information With Devices

Play: 1. Mobile phone | 2. Radio | 3. Television | 4. Email | 5. Video call

Challenge: 1. Mobile phone | 2. Email | 3. Mobile phone | 4. Television | 5. Television | 6. Video call | 7. Radio | 8. Radio | 9. Video call | 10. Email

Lesson 16 — Calculating With Simple Data

Play: 1. Count | 2. Sum | 3. Average | 4. Maximum | 5. Minimum

Challenge: 1. Average | 2. Minimum | 3. Count | 4. Sum | 5. Maximum | 6. Average | 7. Maximum | 8. Count | 9. Minimum | 10. Sum

Lesson 17 — Steps in Data Collection

Play: 1. Research question | 2. Sample | 3. Consent | 4. Tally | 5. Conclusion

Challenge: 1. Consent | 2. Sample | 3. Research question | 4. Consent | 5. Tally | 6. Research question | 7. Sample | 8. Conclusion | 9. Tally | 10. Conclusion

Lesson 18 — Data Collection Tools

Play: 1. Interview | 2. Photograph | 3. Newspaper | 4. Questionnaire | 5. Recording

Challenge: 1. Questionnaire | 2. Newspaper | 3. Newspaper | 4. Recording | 5. Interview | 6. Recording | 7. Photograph | 8. Questionnaire | 9. Interview | 10. Photograph

Lesson 19 — Sketch a Simple Community Map

Play: 1. Map | 2. Symbol | 3. Key | 4. Direction | 5. Scale

Challenge: 1. Direction | 2. Symbol | 3. Symbol | 4. Scale | 5. Key | 6. Key | 7. Scale | 8. Map | 9. Direction | 10. Map

Lesson 20 — Digital Tools in the Community

Play: 1. School | 2. Clinic | 3. Shop | 4. Farm | 5. Home

Challenge: 1. Farm | 2. Shop | 3. Farm | 4. School | 5. Home | 6. School | 7. Home | 8. Clinic | 9. Shop | 10. Clinic

Lesson 21 — Energy-Saving Technology

Play: 1. Energy-efficient bulb | 2. Switch off | 3. Solar charger | 4. Proper maintenance | 5. Reuse

Challenge: 1. Switch off | 2. Solar charger | 3. Reuse | 4. Switch off | 5. Proper maintenance | 6. Energy-efficient bulb | 7. Reuse | 8. Proper maintenance | 9. Solar charger | 10. Energy-efficient bulb

Lesson 22 — Digital Devices and Peripherals

Play: 1. Input device | 2. Output device | 3. Storage device | 4. Processor | 5. Peripheral

Challenge: 1. Input device | 2. Output device | 3. Processor | 4. Storage device | 5. Output device | 6. Processor | 7. Peripheral | 8. Storage device | 9. Input device | 10. Peripheral

Lesson 23 — Changing Technology Around Us

Play: 1. Benefit | 2. Risk | 3. Healthcare | 4. Farming | 5. Privacy

Challenge: 1. Healthcare | 2. Risk | 3. Risk | 4. Farming | 5. Privacy | 6. Benefit | 7. Benefit | 8. Healthcare | 9. Privacy | 10. Farming

Lesson 24 — PowerPoint Home Tools

Play: 1. Slide | 2. Theme | 3. Transition | 4. Animation | 5. Presenter

Challenge: 1. Theme | 2. Presenter | 3. Animation | 4. Presenter | 5. Transition | 6. Animation | 7. Transition | 8. Slide | 9. Slide | 10. Theme

Lesson 25 — Make a Three-Slide Presentation

Play: 1. Opening slide | 2. Main slide | 3. Image | 4. Closing slide | 5. Rehearsal

Challenge: 1. Image | 2. Closing slide | 3. Rehearsal | 4. Opening slide | 5. Main slide | 6. Image | 7. Rehearsal | 8. Main slide | 9. Opening slide | 10. Closing slide

Lesson 26 — Word Processor Home Tools

Play: 1. Word processor | 2. Bold | 3. Alignment | 4. Clipboard | 5. Spell check

Challenge: 1. Bold | 2. Spell check | 3. Clipboard | 4. Bold | 5. Word processor | 6. Alignment | 7. Word processor | 8. Spell check | 9. Alignment | 10. Clipboard

Lesson 27 — Format a Simple Document

Play: 1. Word processor | 2. Bold | 3. Alignment | 4. Clipboard | 5. Spell check

Challenge: 1. Bold | 2. Alignment | 3. Word processor | 4. Clipboard | 5. Bold | 6. Word processor | 7. Alignment | 8. Spell check | 9. Spell check | 10. Clipboard